Nassau River-St. Johns River Marshes Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

Contents

Indicators

Nutrients

Total Nitrogen - Discrete

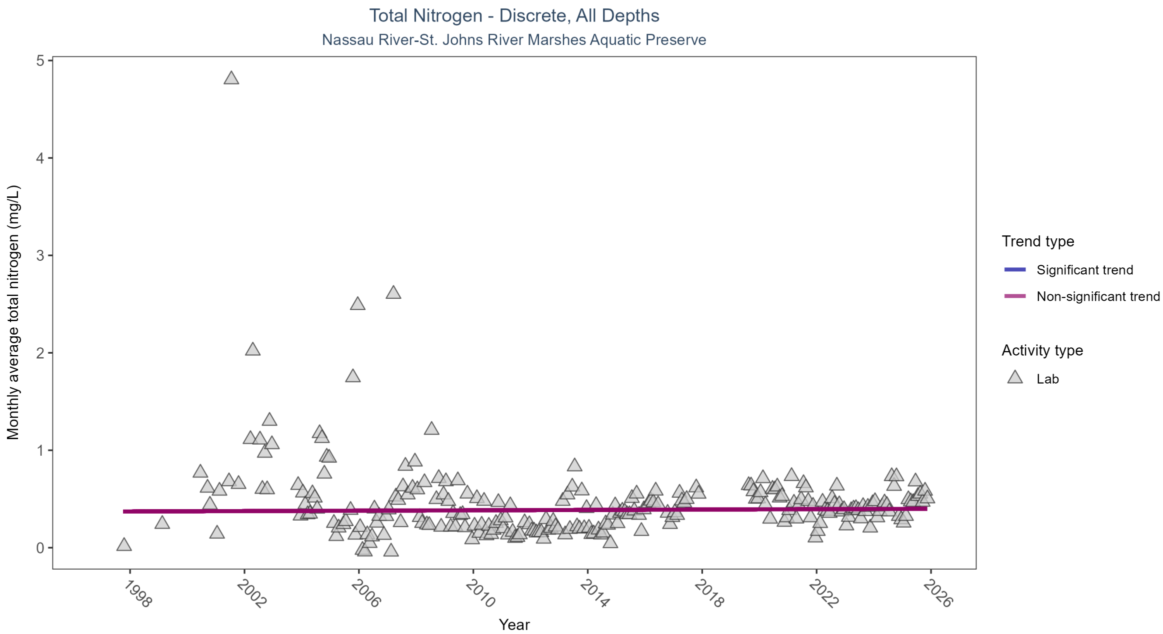

Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Total nitrogen showed no detectable trend between 1997 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1114	27	1997 - 2025	0.41405	0.02443	0.37083	0.00098	0.6327

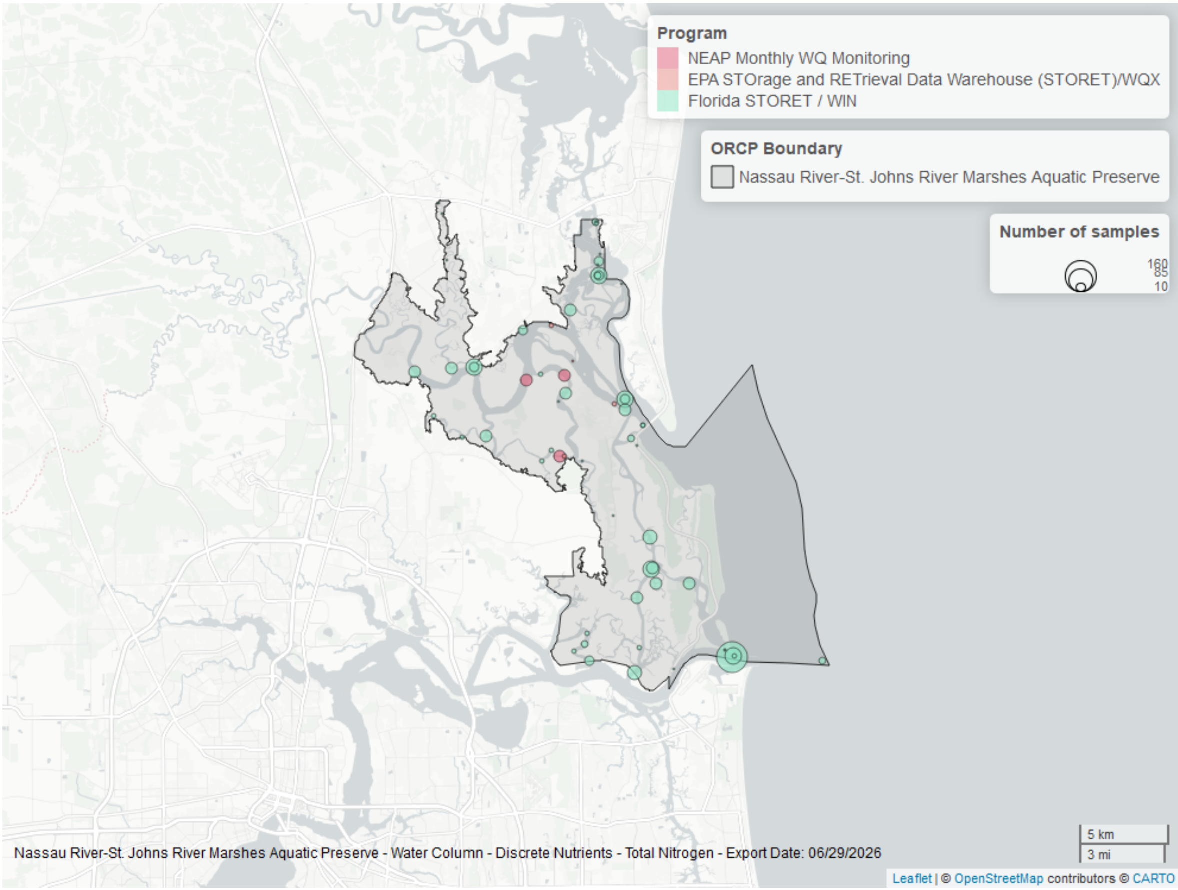

Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

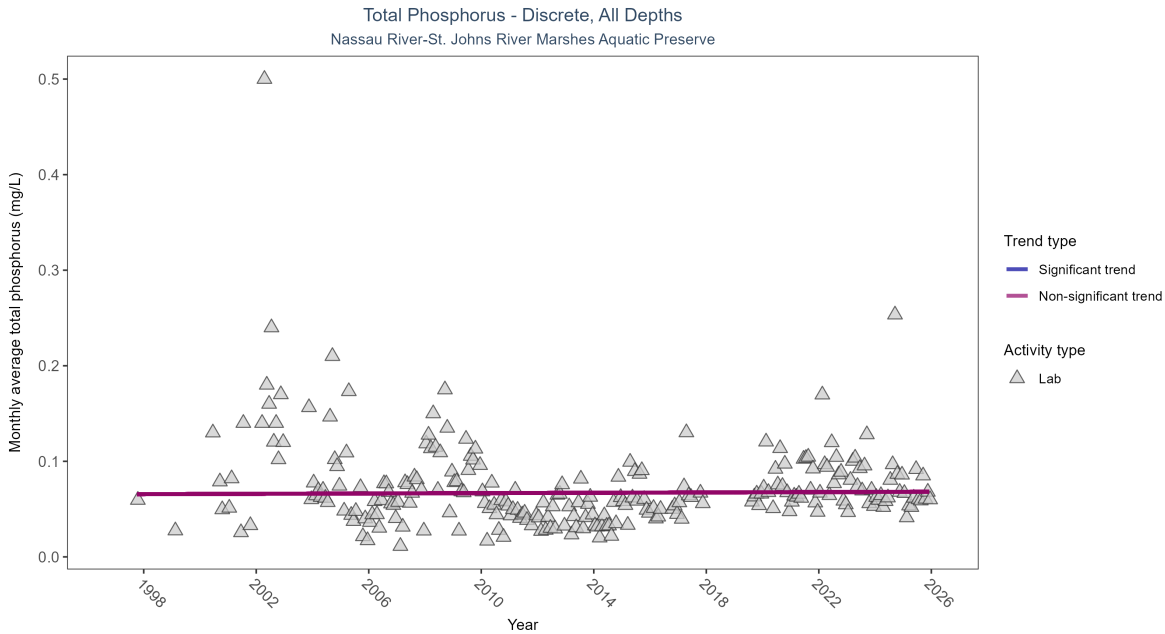

Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1997 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	1676	27	1997 - 2025	0.06625	0.01212	0.06554	0.00009	0.7119

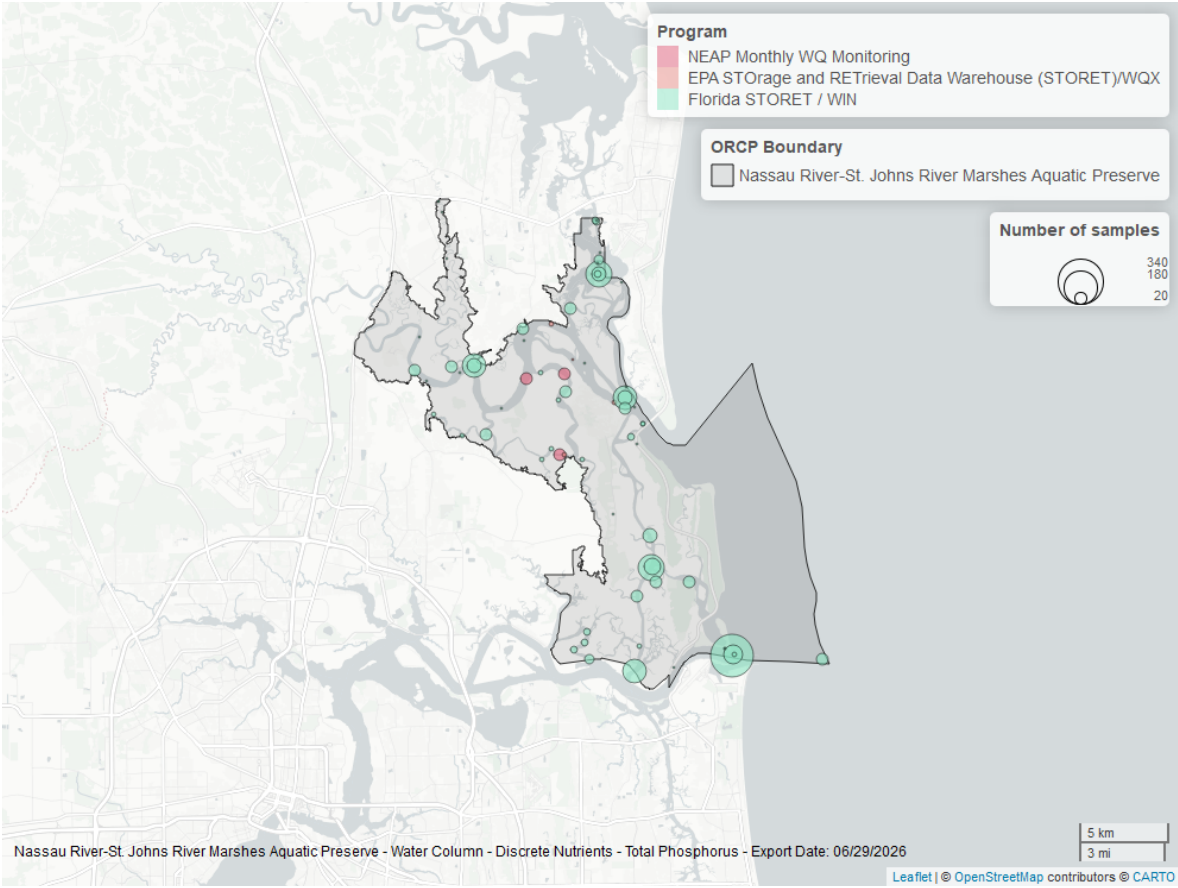

Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

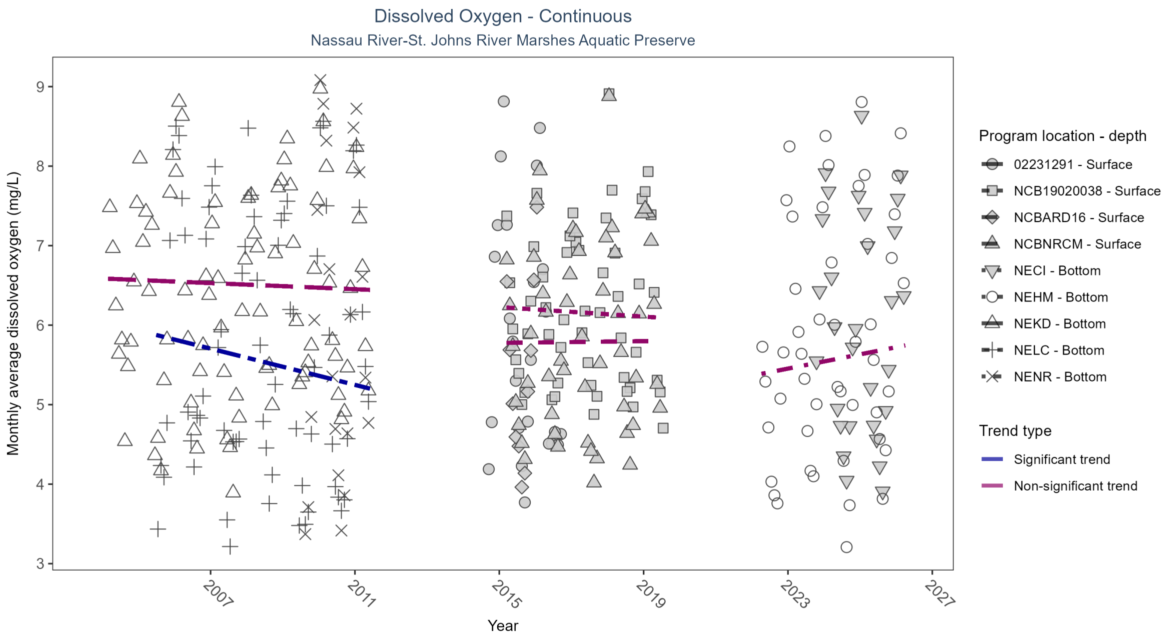

Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: At one program location, monthly average dissolved oxygen decreased by 0.11 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at four locations. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02231291	Insufficient data	689	3	2014 - 2016	5.70	-	-	-	-
NECI	Insufficient data	77240	4	2023 - 2026	6.20	-	-	-	-
NEHM	No detectable trend	127898	5	2022 - 2026	5.70	0.22	5.36	0.09	0.14
NEKD	No detectable trend	110958	8	2004 - 2011	6.40	-0.04	6.59	-0.02	0.56
NELC	Decreasing trend	95860	7	2005 - 2011	5.60	-0.34	5.93	-0.11	0
NENR	Insufficient data	31438	3	2009 - 2011	5.80	-	-	-	-
NCB19020038	No detectable trend	34476	5	2015 - 2019	6.16	-0.03	6.23	-0.03	0.61
NCBARD16	Insufficient data	7417	2	2015 - 2016	5.08	-	-	-	-
NCBNRCM	No detectable trend	35477	5	2015 - 2019	5.79	0.05	5.78	0.01	0.8

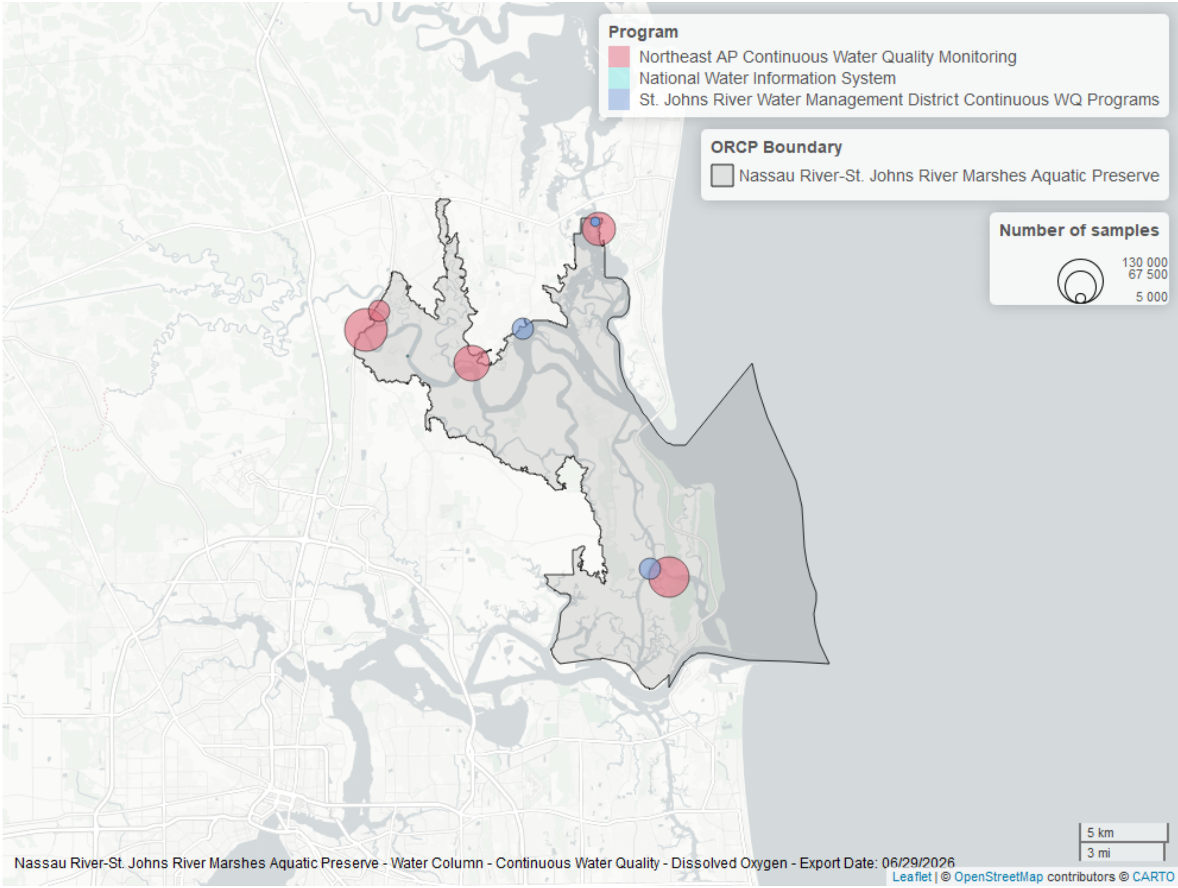

Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

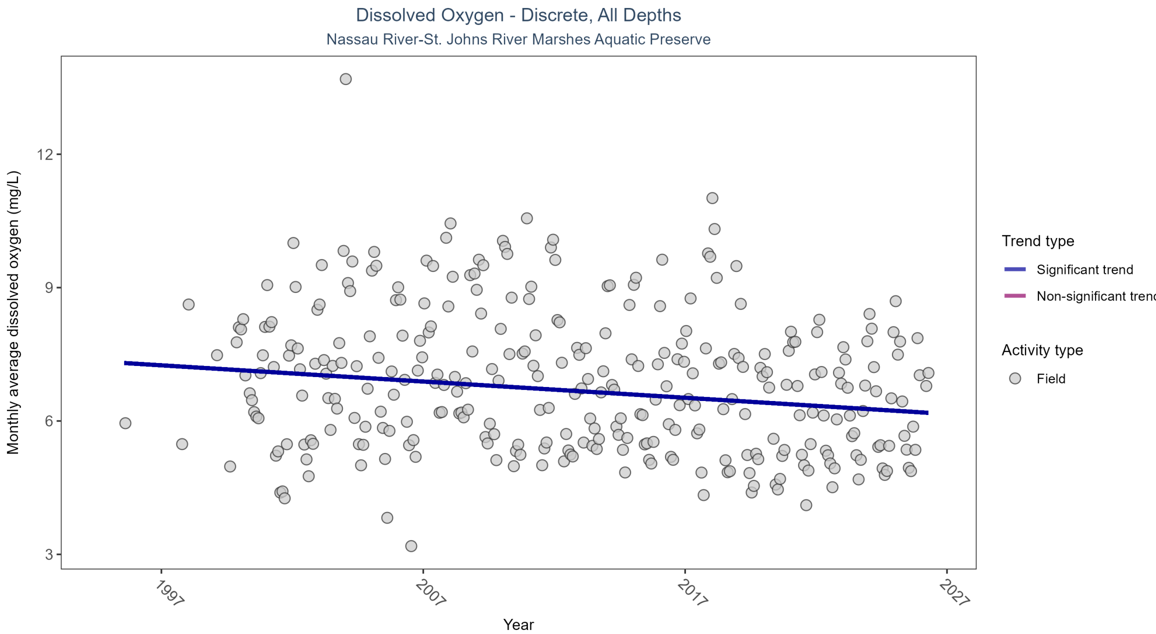

Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Monthly average dissolved oxygen decreased by 0.04 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	22738	31	1995 - 2026	6.8	-0.27173	7.32517	-0.03647	0

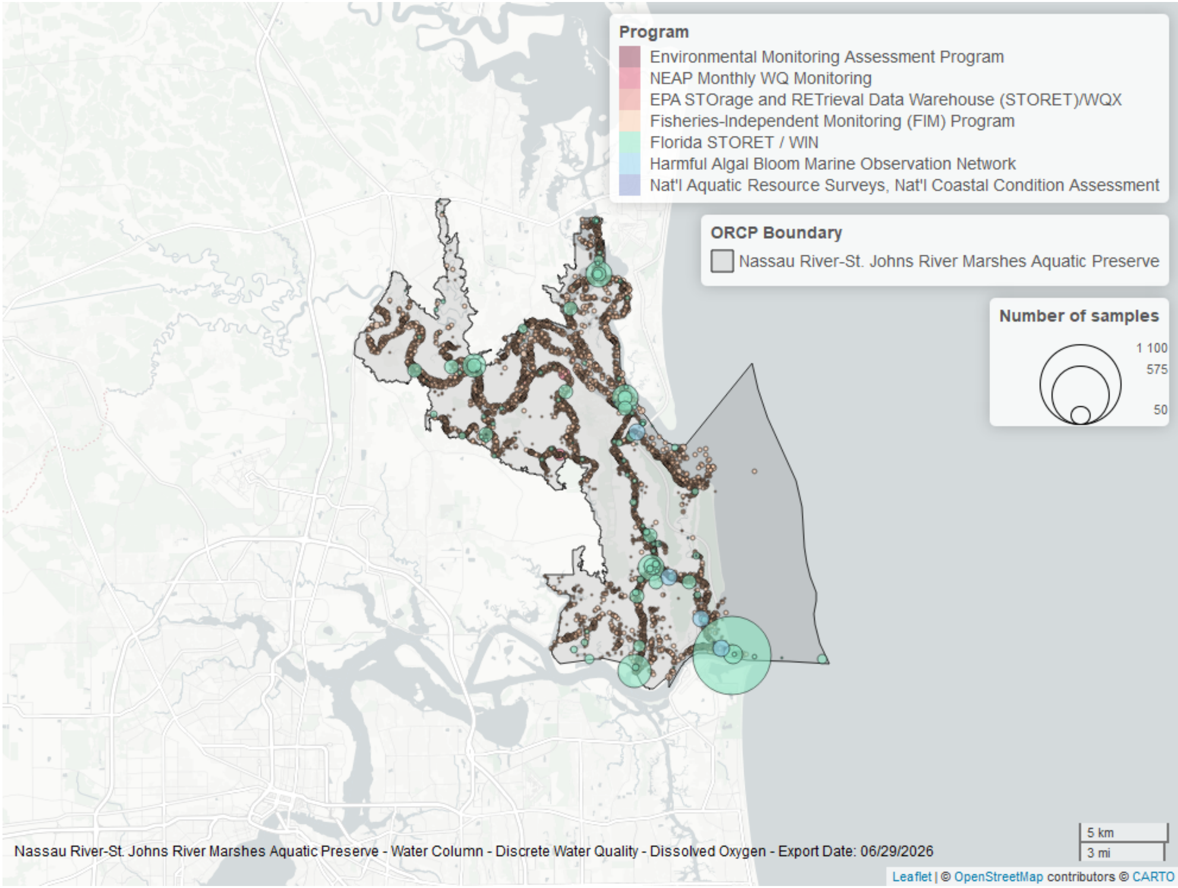

Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

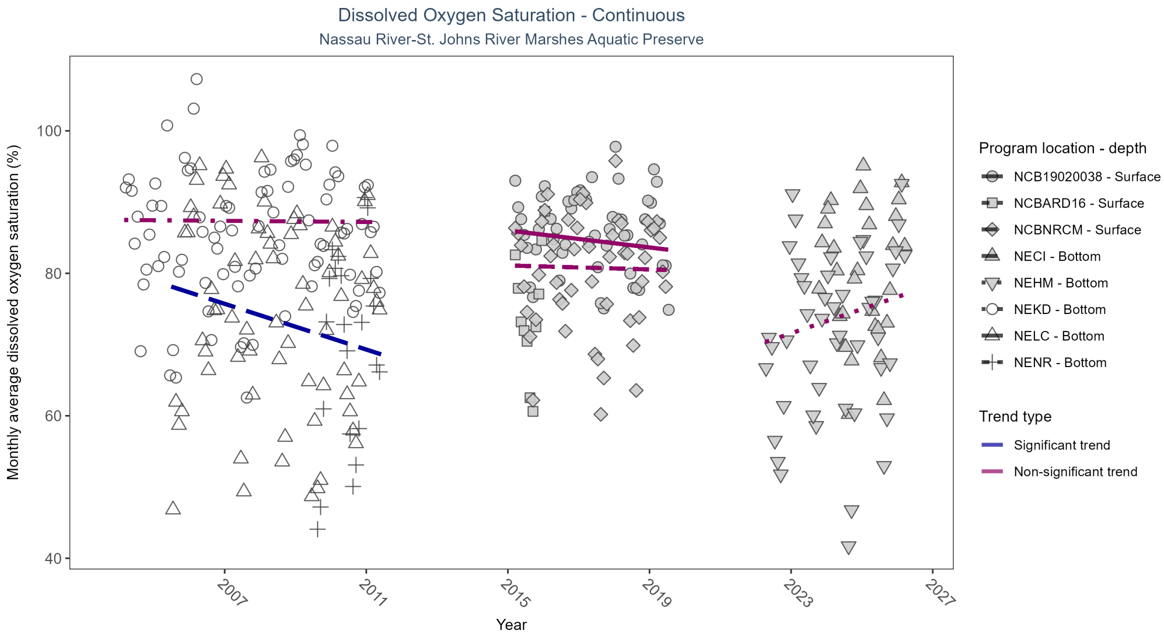

Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

\begin{table}[H] \caption{At one program location, monthly average dissolved oxygen saturation decreased by 1.60% per year. No detectable change in monthly average dissolved oxygen saturation was observed at four locations. There was insufficient data to fit a model for three locations.}

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data	77587	4	2023 - 2026	82.50	-	-	-	-
NEHM	No detectable trend	130580	5	2022 - 2026	72.90	0.19	69.88	1.7	0.2
NEKD	No detectable trend	110983	8	2004 - 2011	89.10	0	87.49	-0.04	0.93
NELC	Decreasing trend	95868	7	2005 - 2011	77.90	-0.27	78.93	-1.6	0.01
NENR	Insufficient data	31438	3	2009 - 2011	73.20	-	-	-	-
NCB19020038	No detectable trend	34438	5	2015 - 2019	87.06	-0.16	86.02	-0.6	0.13
NCBARD16	Insufficient data	7417	2	2015 - 2016	74.44	-	-	-	-
NCBNRCM	No detectable trend	35240	5	2015 - 2019	82.06	-0.05	81.1	-0.14	0.8

\end{table}

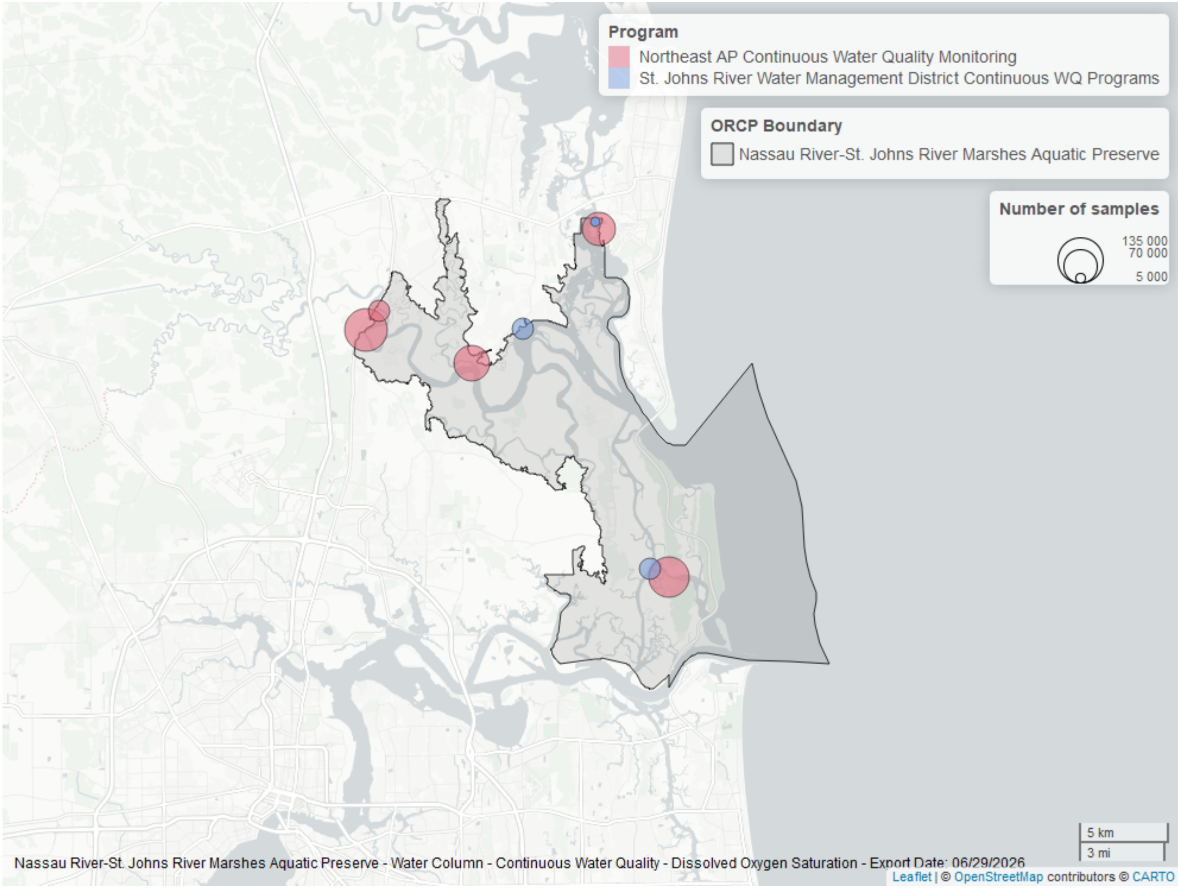

Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

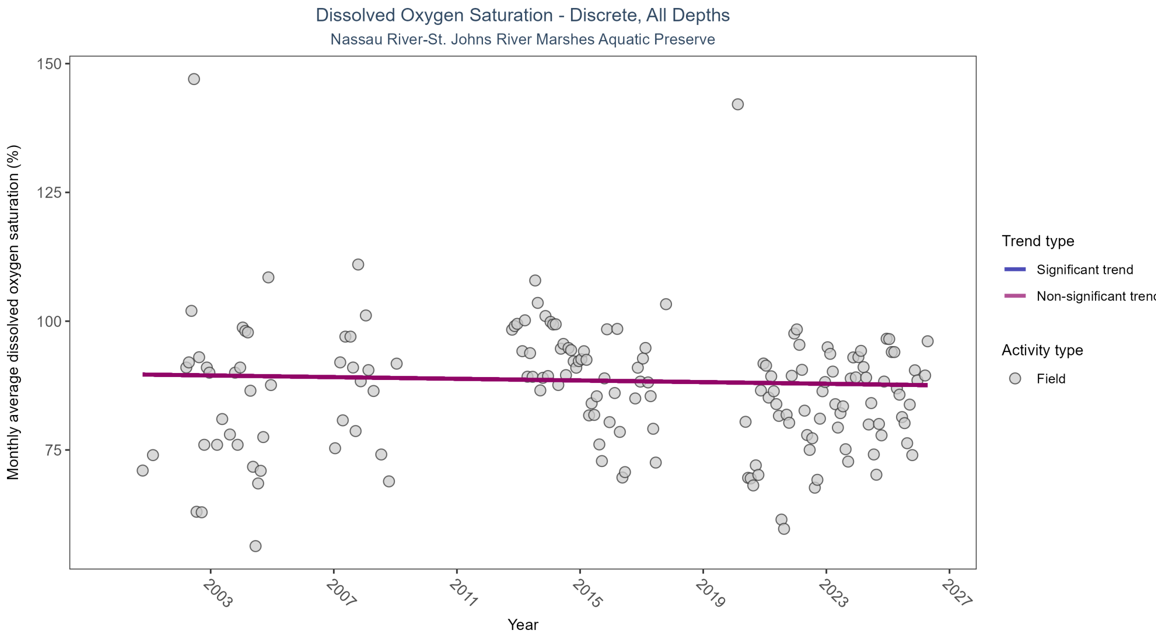

Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 5: Dissolved oxygen saturation showed no detectable trend between 2000 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	1521	21	2000 - 2026	88	-0.07841	89.70324	-0.08082	0.1518

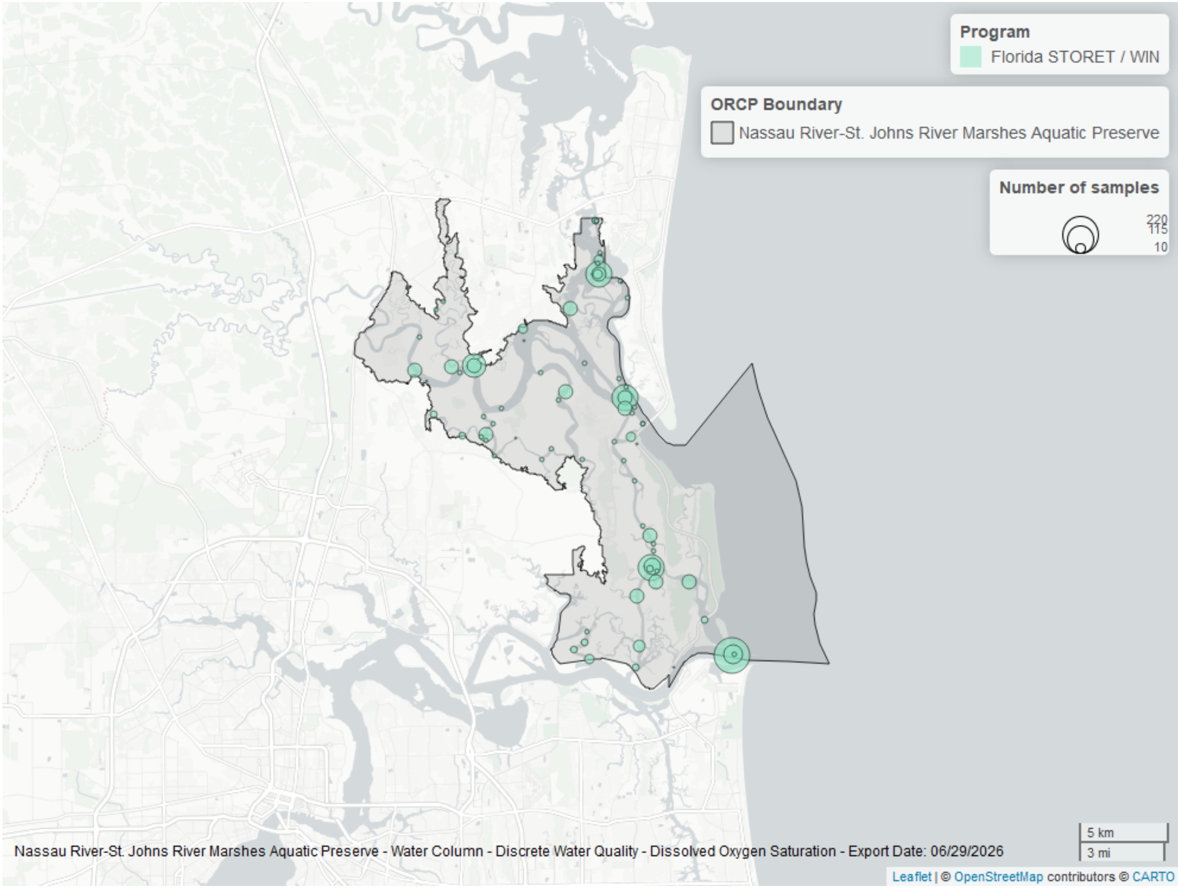

Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

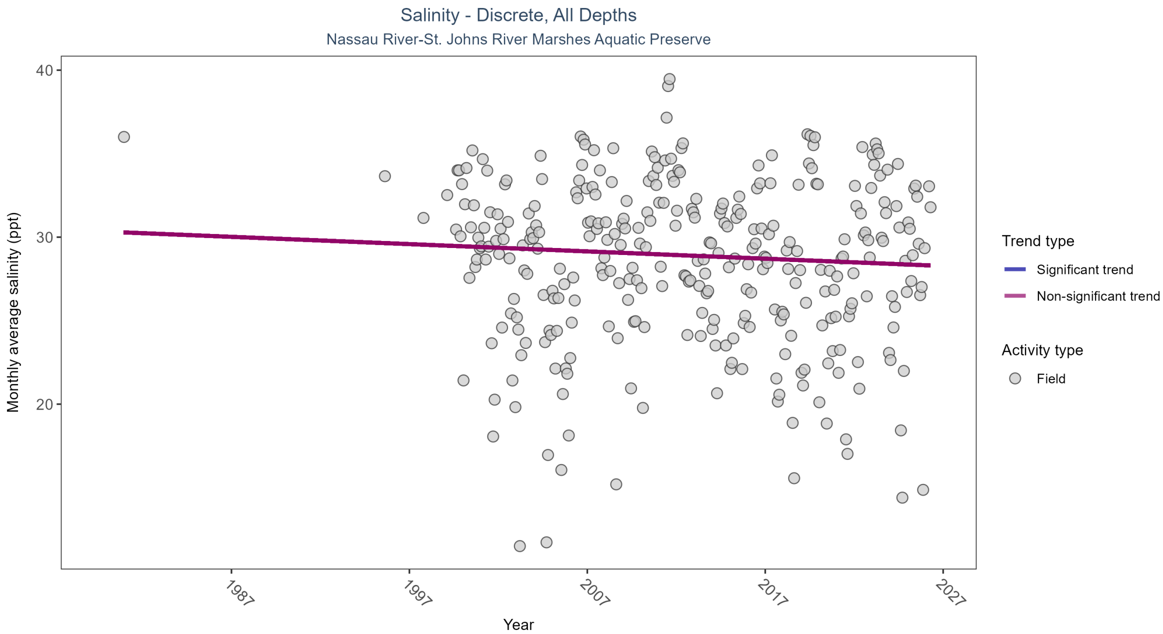

Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 6: Salinity showed no detectable trend between 1980 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	22844	31	1980 - 2026	30.5	-0.04983	30.32475	-0.04363	0.1904

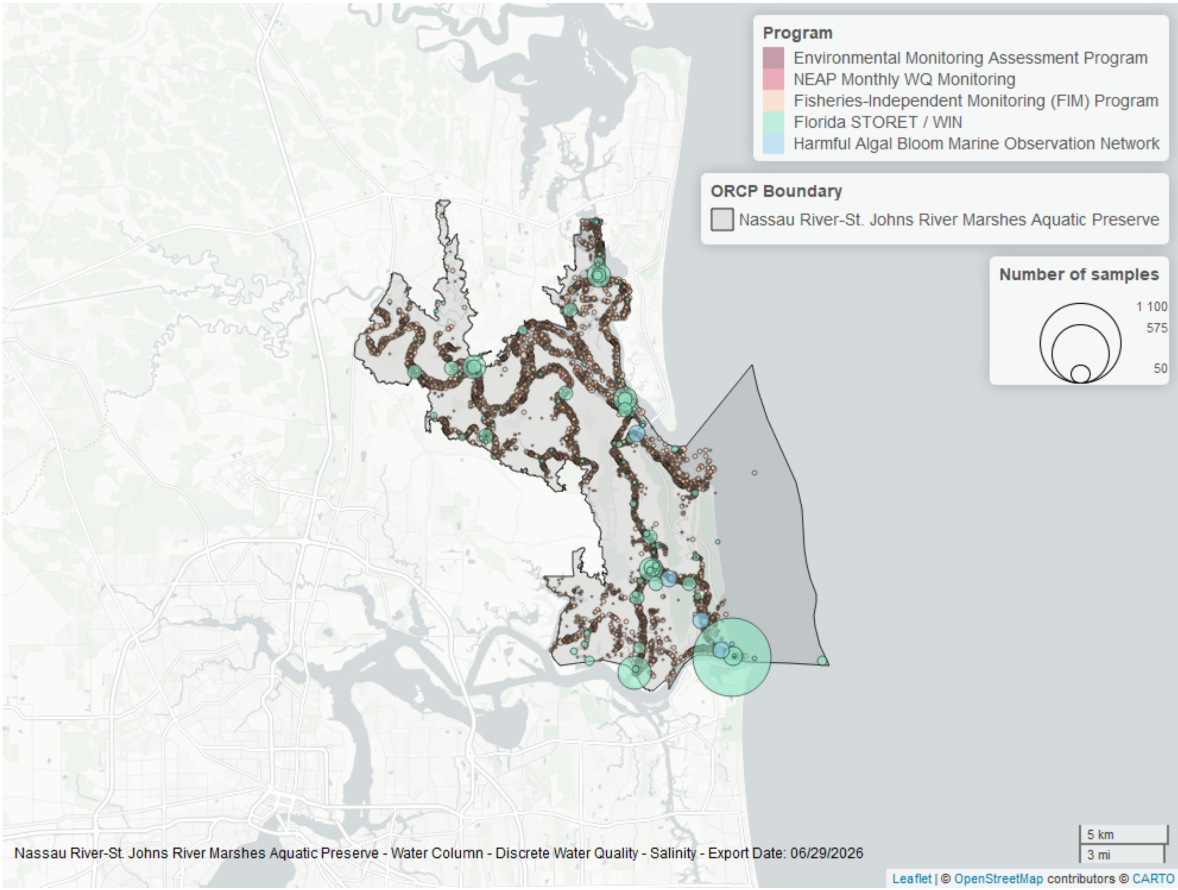

Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

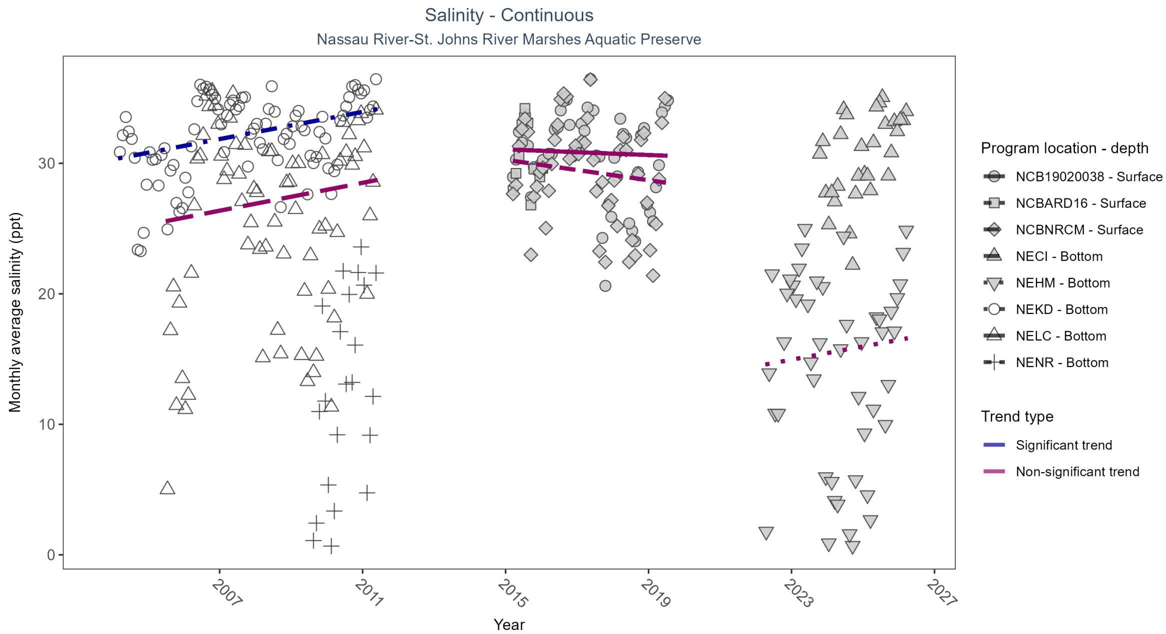

Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 7: At one program location, monthly average salinity increased by 0.52 ppt per year. No detectable change in monthly average salinity was observed at four locations. There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data	76803	4	2023 - 2026	30.80	-	-	-	-
NEHM	No detectable trend	129295	5	2022 - 2026	15.60	0.04	14.45	0.51	0.84
NEKD	Increasing trend	118328	8	2004 - 2011	33.20	0.31	30.29	0.52	0
NELC	No detectable trend	100339	7	2005 - 2011	27.90	0.09	25.3	0.53	0.44
NENR	Insufficient data	31438	3	2009 - 2011	12.40	-	-	-	-
NCB19020038	No detectable trend	34438	5	2015 - 2019	31.75	-0.06	31.07	-0.11	1
NCBARD16	Insufficient data	7418	2	2015 - 2016	30.07	-	-	-	-
NCBNRCM	No detectable trend	35411	5	2015 - 2019	30.29	-0.12	30.26	-0.39	0.55

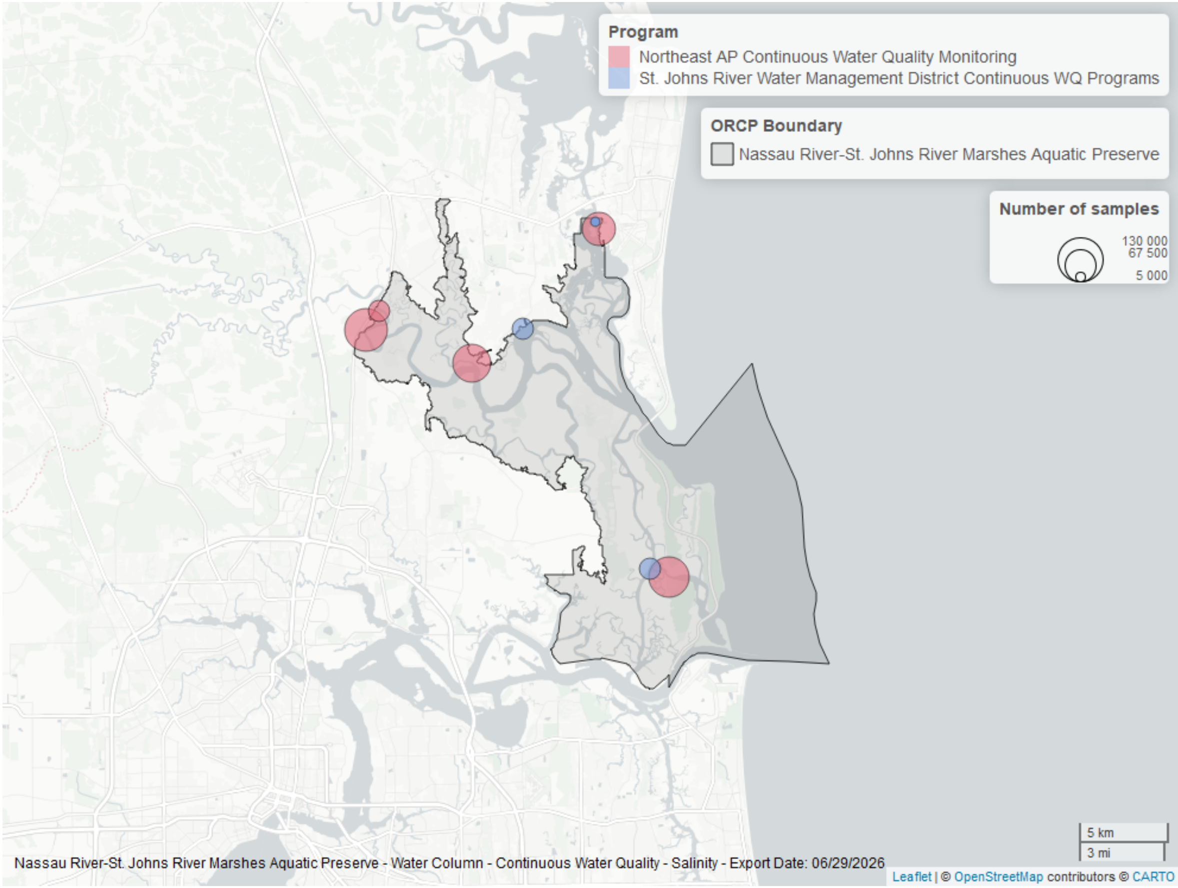

Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

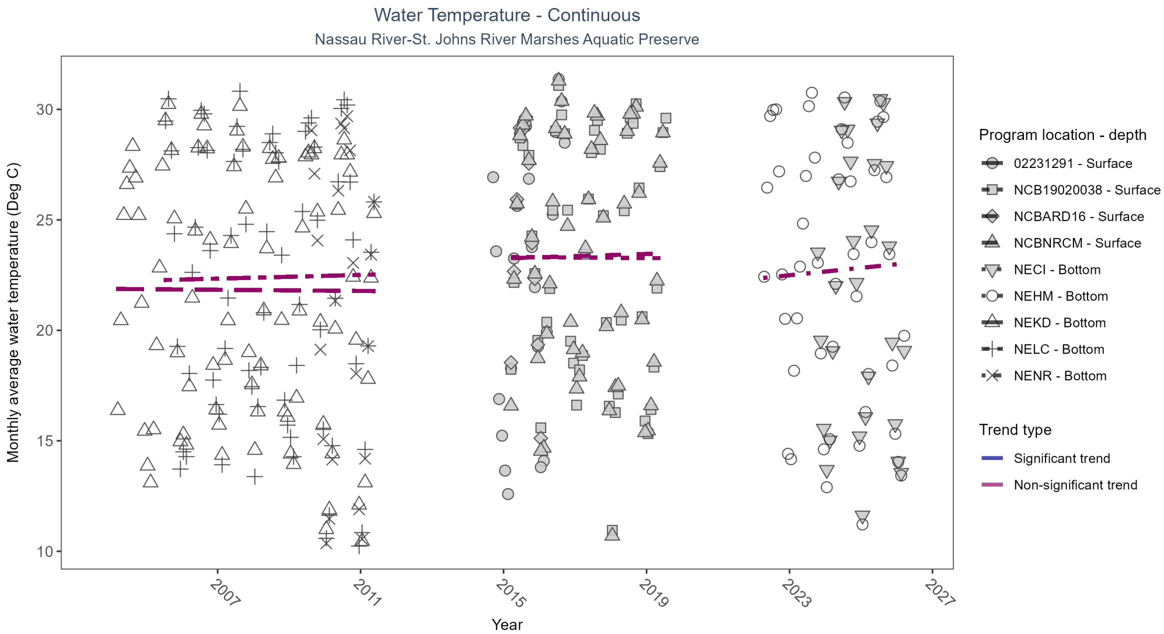

Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: No detectable change in monthly average water temperature was observed at five locations. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02231291	Insufficient data	710	3	2014 - 2016	23.50	-	-	-	-
NECI	Insufficient data	80058	4	2023 - 2026	20.70	-	-	-	-
NEHM	No detectable trend	135834	5	2022 - 2026	23.20	0.08	22.33	0.17	0.62
NEKD	No detectable trend	118328	8	2004 - 2011	22.10	-0.04	21.88	-0.01	0.81
NELC	No detectable trend	100343	7	2005 - 2011	22.70	0.05	22.26	0.04	0.74
NENR	Insufficient data	31438	3	2009 - 2011	22.00	-	-	-	-
NCB19020038	No detectable trend	34483	5	2015 - 2019	24.78	-0.03	23.32	-0.01	1
NCBARD16	Insufficient data	7419	2	2015 - 2016	25.08	-	-	-	-
NCBNRCM	No detectable trend	35817	5	2015 - 2019	24.03	0.07	23.26	0.05	0.67

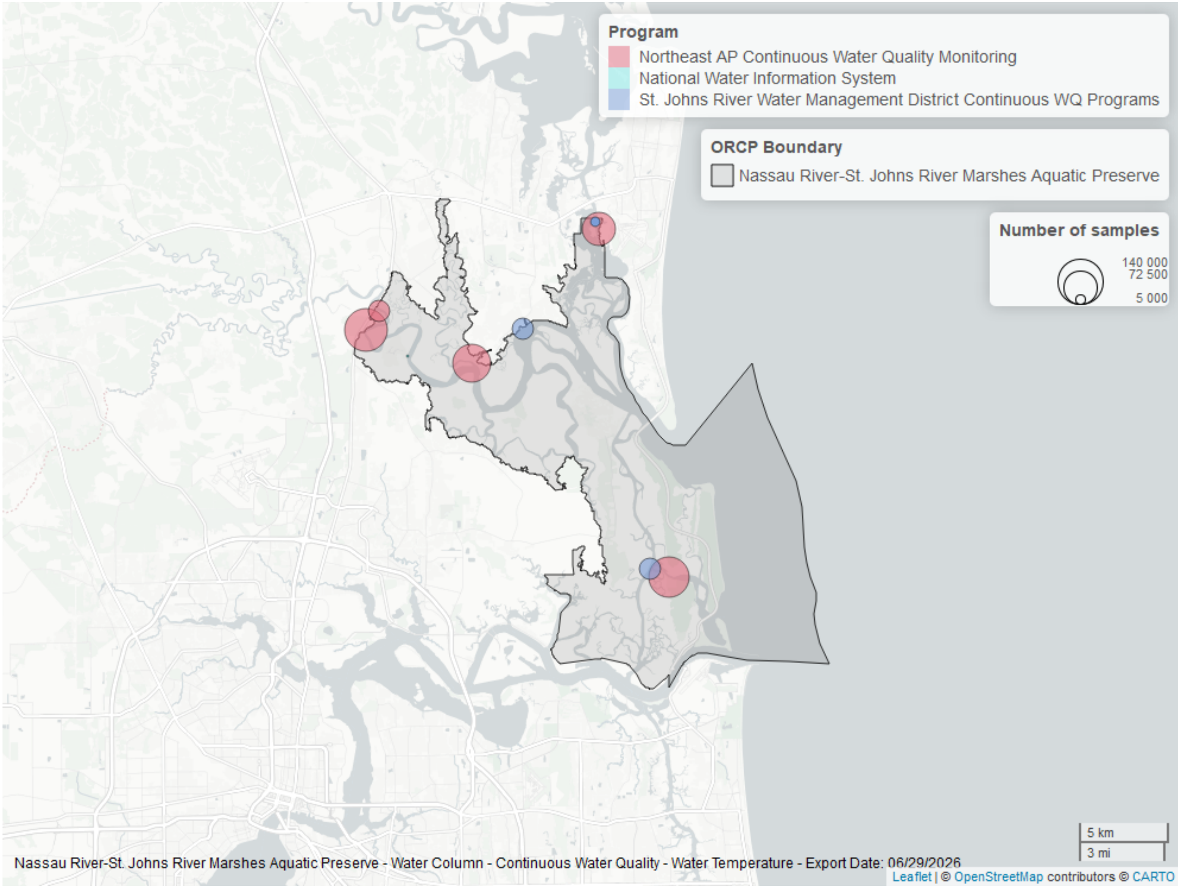

Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve.* The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

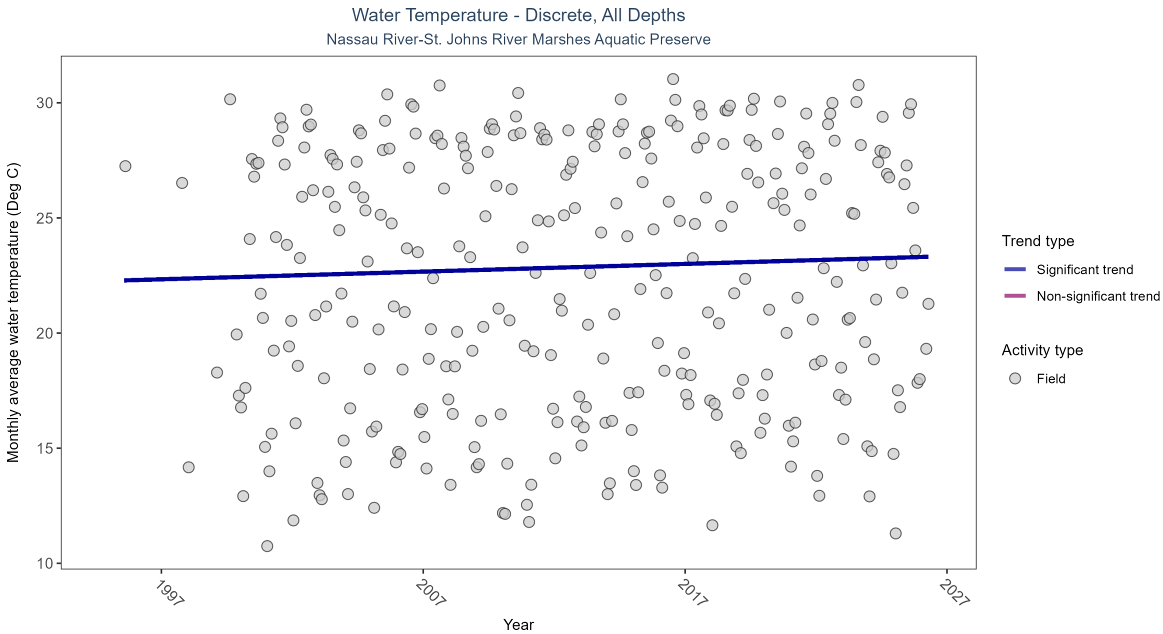

Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 9: Monthly average water temperature increased by 0.03Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	22906	31	1995 - 2026	23	0.12295	22.26829	0.03339	0.0022

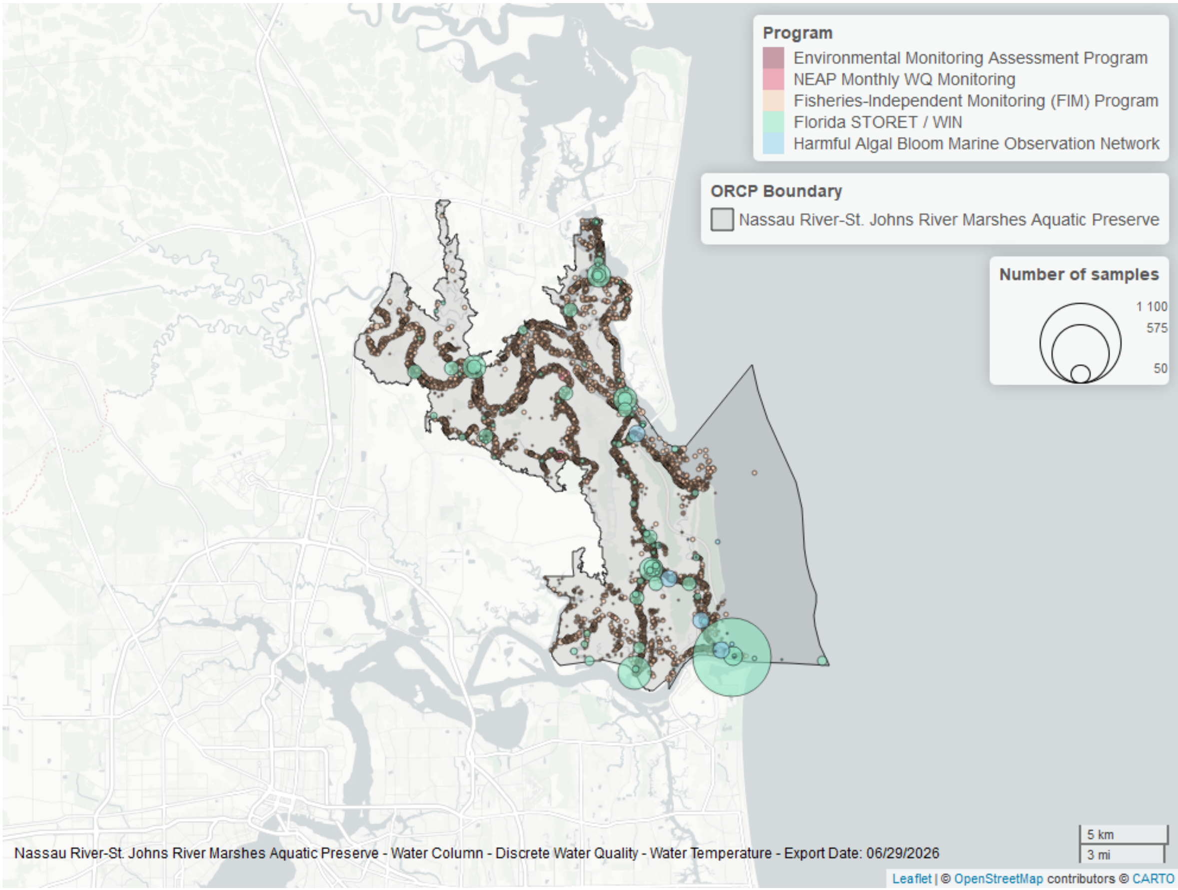

Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

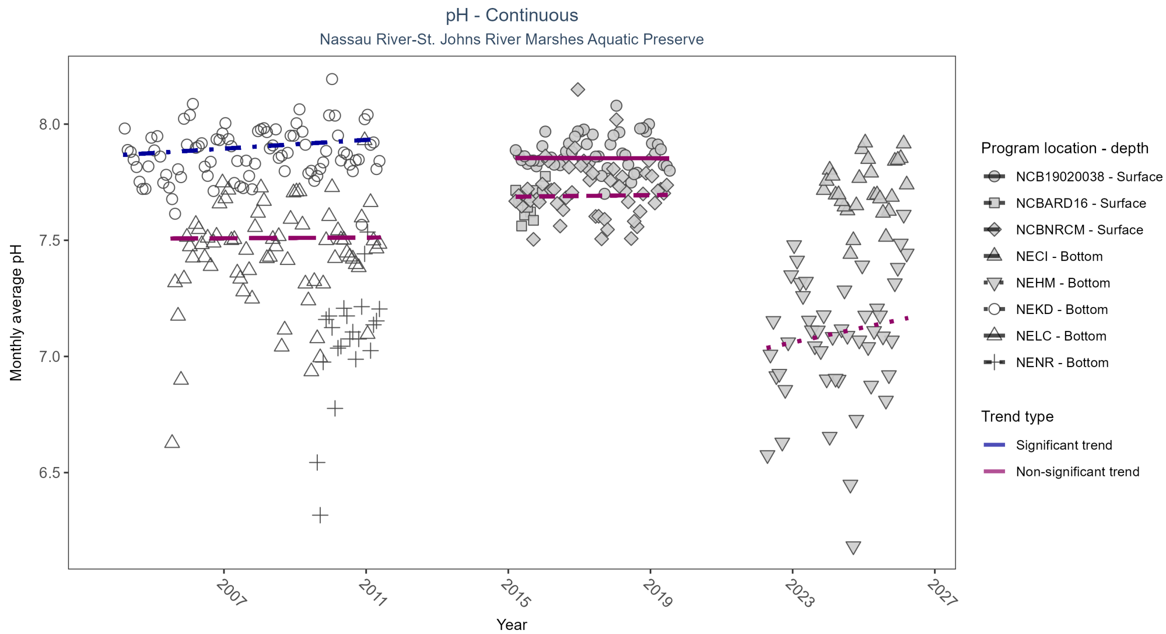

Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 10: At one program location, monthly average pH increased by 0.01 pH units per year. No detectable change in monthly average pH was observed at four locations. There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data	77746	4	2023 - 2026	7.70	-	-	-	-
NEHM	No detectable trend	133897	5	2022 - 2026	7.10	0.22	7.03	0.03	0.14
NEKD	Increasing trend	113471	8	2004 - 2011	7.90	0.19	7.87	0.01	0.05
NELC	No detectable trend	96488	7	2005 - 2011	7.50	0.02	7.51	0	0.91
NENR	Insufficient data	31438	3	2009 - 2011	7.10	-	-	-	-
NCB19020038	No detectable trend	34405	5	2015 - 2019	7.88	-0.01	7.85	0	1
NCBARD16	Insufficient data	6952	2	2015 - 2016	7.66	-	-	-	-
NCBNRCM	No detectable trend	35819	5	2015 - 2019	7.72	-0.01	7.69	0	0.93

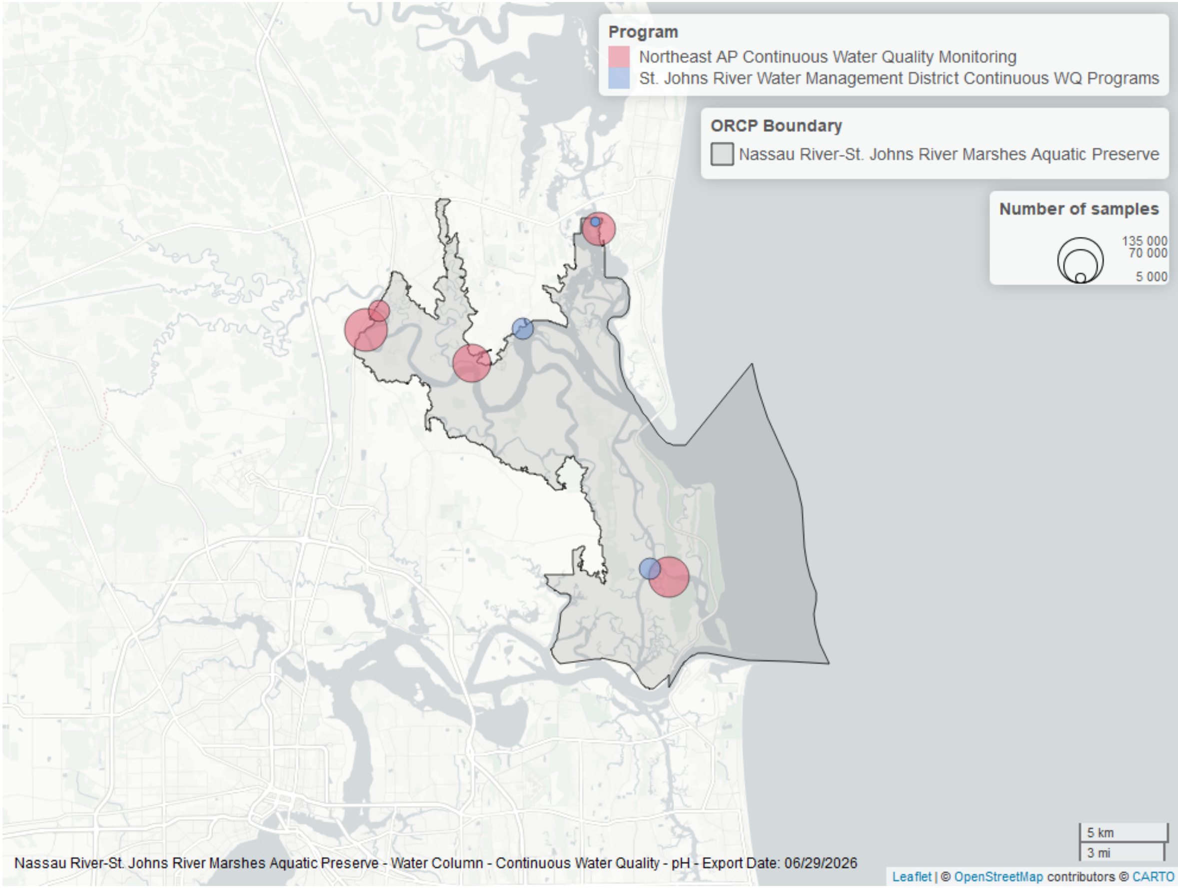

Figure 22: Map showing location of ph continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

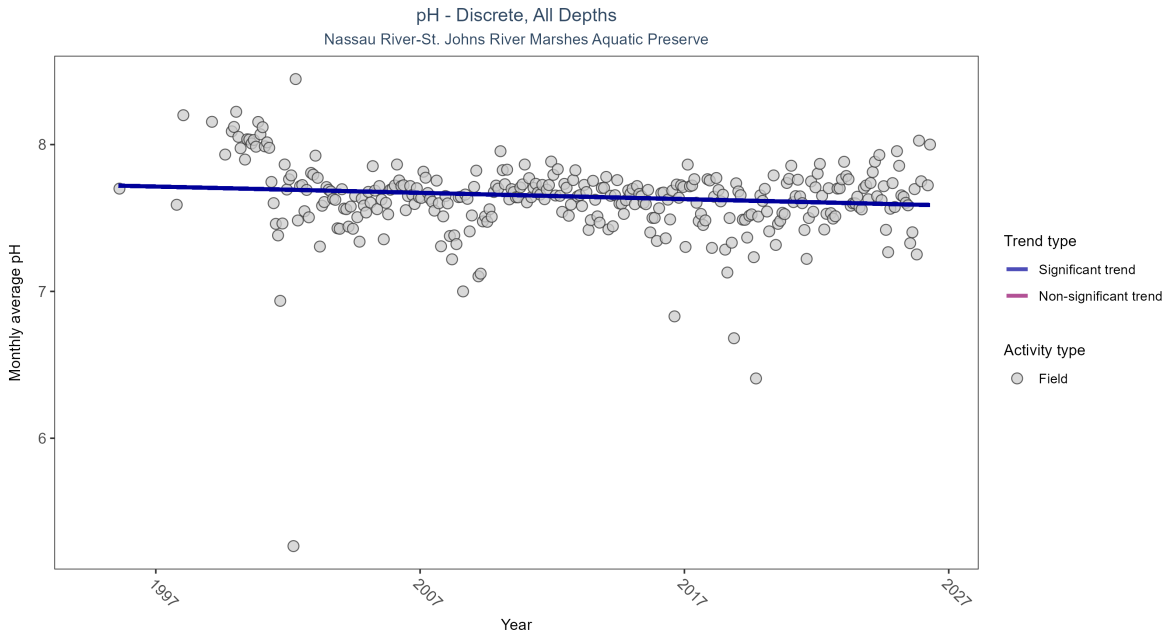

Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 11: Monthly average pH decreased by less than 0.01 pH units per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	22671	31	1995 - 2026	7.6	-0.12673	7.72273	-0.00429	0.0014

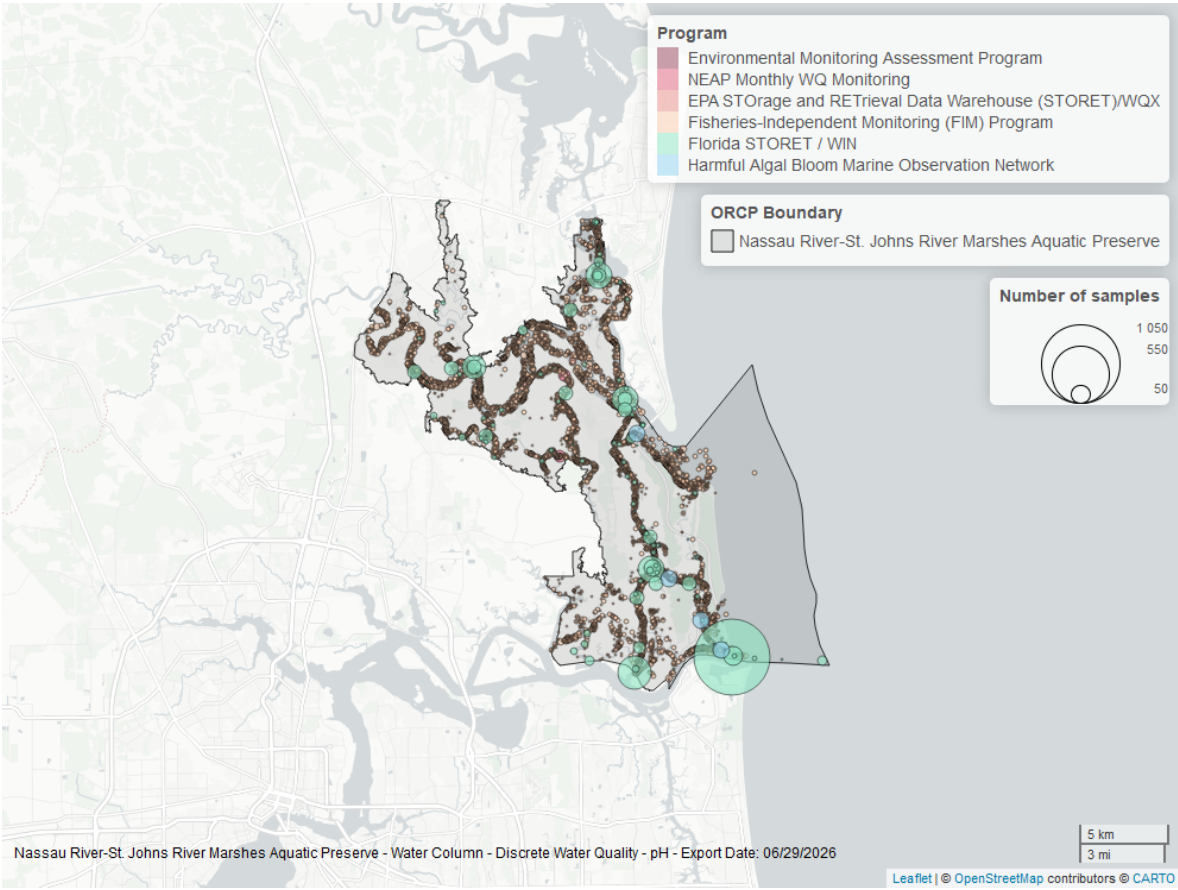

Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

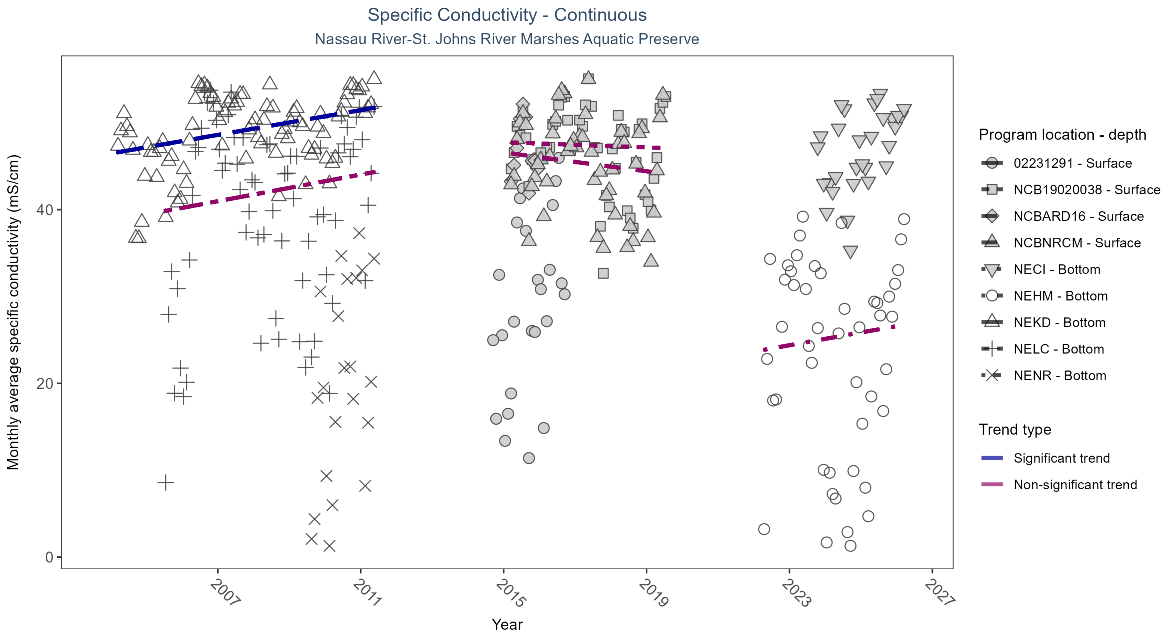

Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 12: At one program location, monthly average specific conductivity increased by 0.71 mS/cm per year. No detectable change in monthly average specific conductivity was observed at four locations. There was insufficient data to fit a model for four locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02231291	Insufficient data	682	3	2014 - 2016	30.40	-	-	-	-
NECI	Insufficient data	76803	4	2023 - 2026	47.29	-	-	-	-
NEHM	No detectable trend	129299	5	2022 - 2026	25.56	0.04	23.65	0.73	0.84
NEKD	Increasing trend	118327	8	2004 - 2011	50.62	0.31	46.46	0.71	0
NELC	No detectable trend	100339	7	2005 - 2011	43.33	0.08	39.43	0.77	0.51
NENR	Insufficient data	31438	3	2009 - 2011	20.04	-	-	-	-
NCB19020038	No detectable trend	34438	5	2015 - 2019	48.67	-0.06	47.75	-0.15	1
NCBARD16	Insufficient data	7418	2	2015 - 2016	46.32	-	-	-	-
NCBNRCM	No detectable trend	35411	5	2015 - 2019	46.65	-0.12	46.58	-0.54	0.55

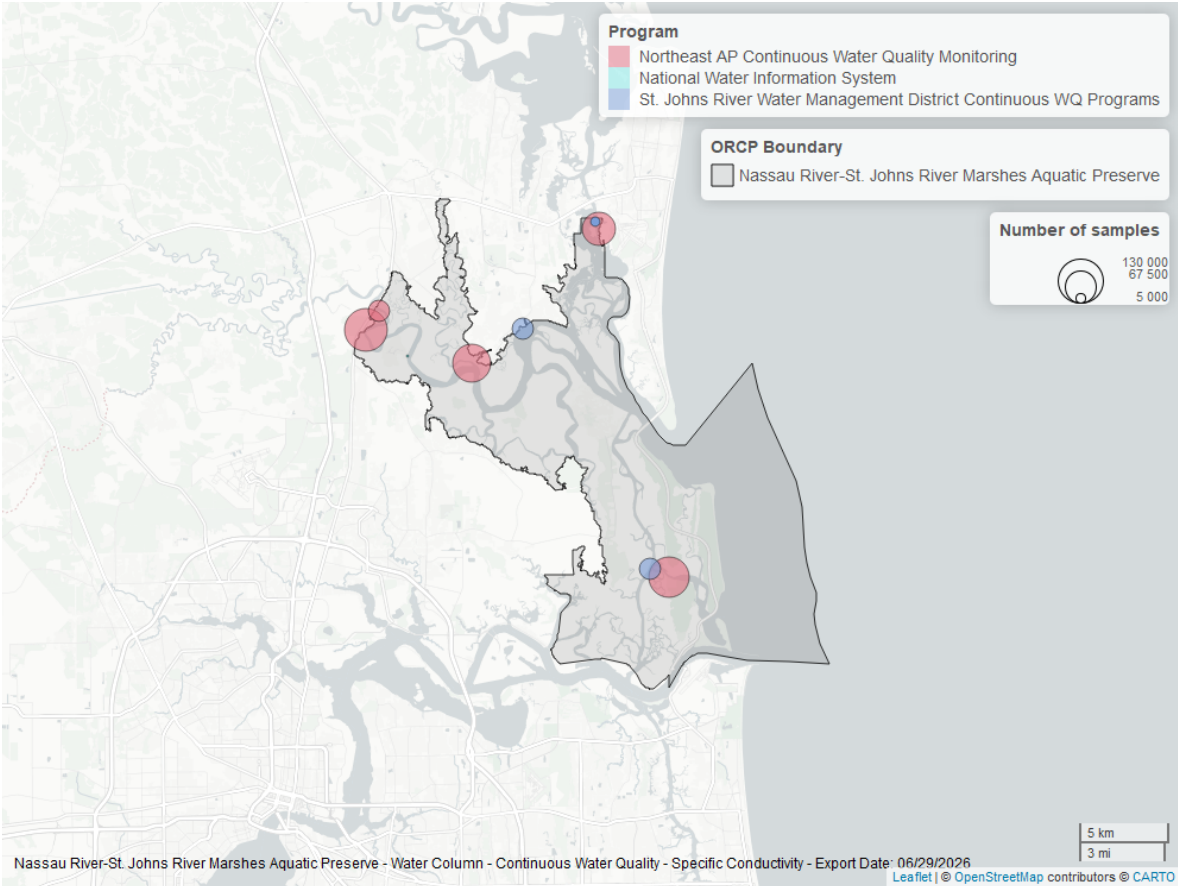

Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

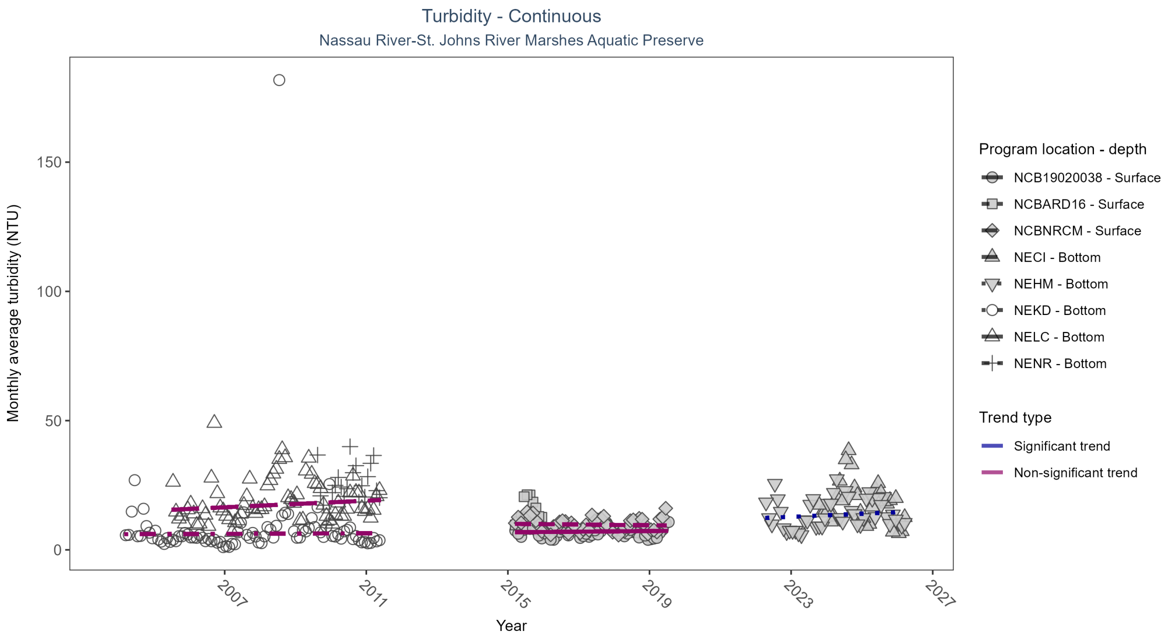

Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: At one program location, monthly average turbidity increased by 0.60 NTU per year. No detectable change in monthly average turbidity was observed at four locations. There was insufficient data to fit a model for three locations.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data	66966	4	2023 - 2026	14.00	-	-	-	-
NEHM	Increasing trend	123949	5	2022 - 2026	12.00	0.36	12.18	0.6	0.01
NEKD	No detectable trend	114181	8	2004 - 2011	4.00	0.02	6.06	0.05	0.87
NELC	No detectable trend	96153	7	2005 - 2011	15.00	0.16	15.18	0.64	0.15
NENR	Insufficient data	31087	3	2009 - 2011	21.00	-	-	-	-
NCB19020038	No detectable trend	31407	5	2015 - 2019	5.77	-0.01	6.79	0.12	0.79
NCBARD16	Insufficient data	7385	2	2015 - 2016	12.89	-	-	-	-
NCBNRCM	No detectable trend	34696	5	2015 - 2019	8.64	-0.1	10.09	-0.14	0.6

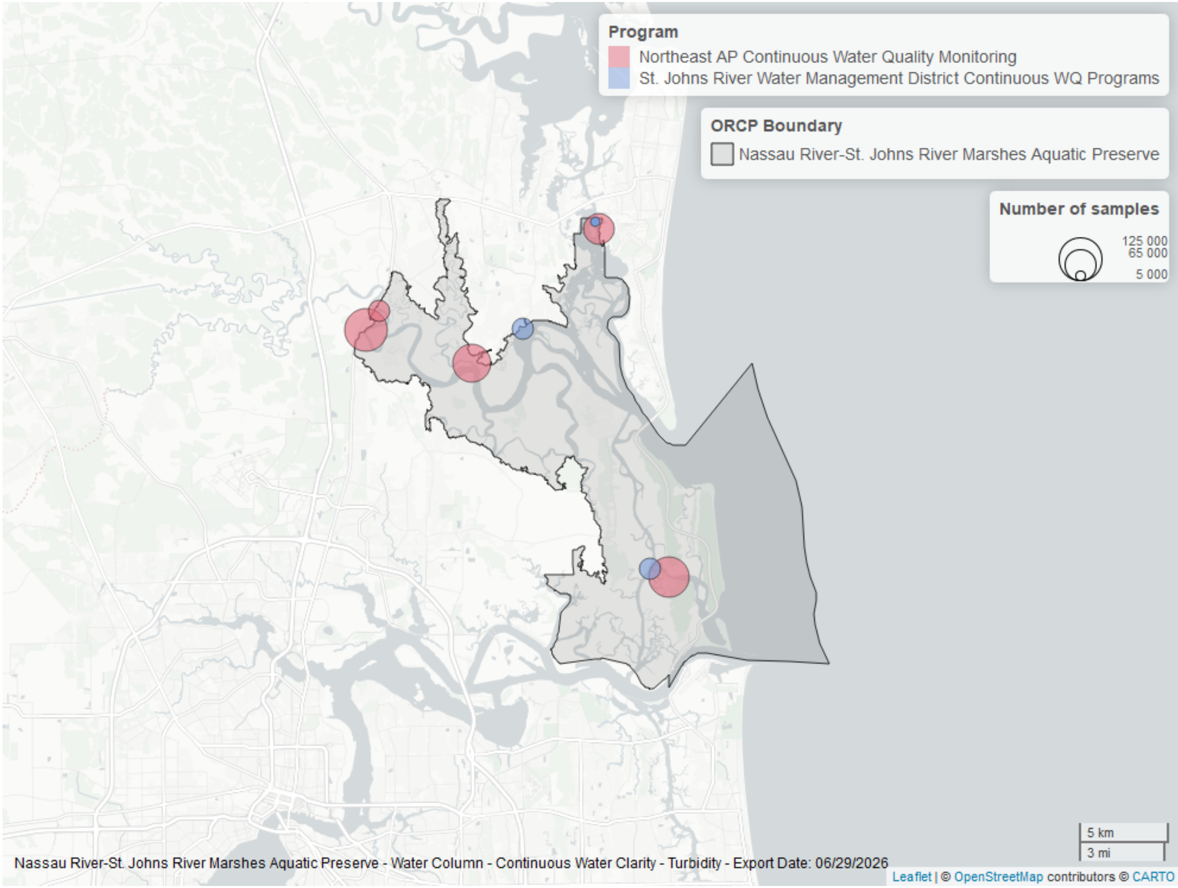

Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

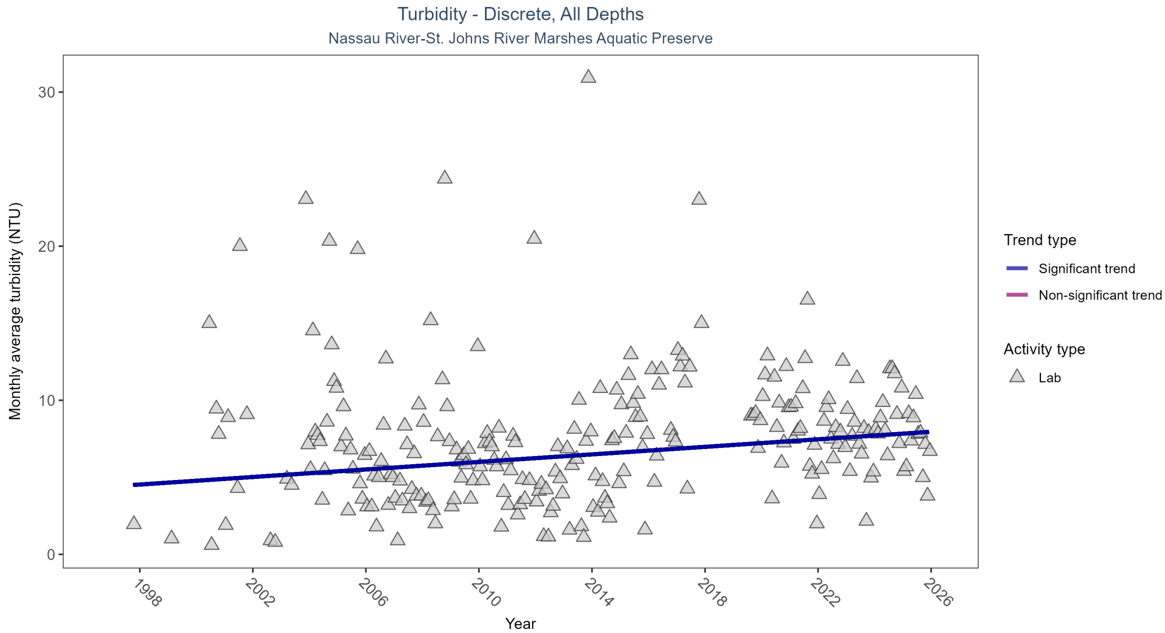

Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 14: Monthly average turbidity increased by 0.12 NTU per year, indicating a decrease in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	1135	27	1997 - 2025	7.1	0.17024	4.40949	0.12225	0.0002

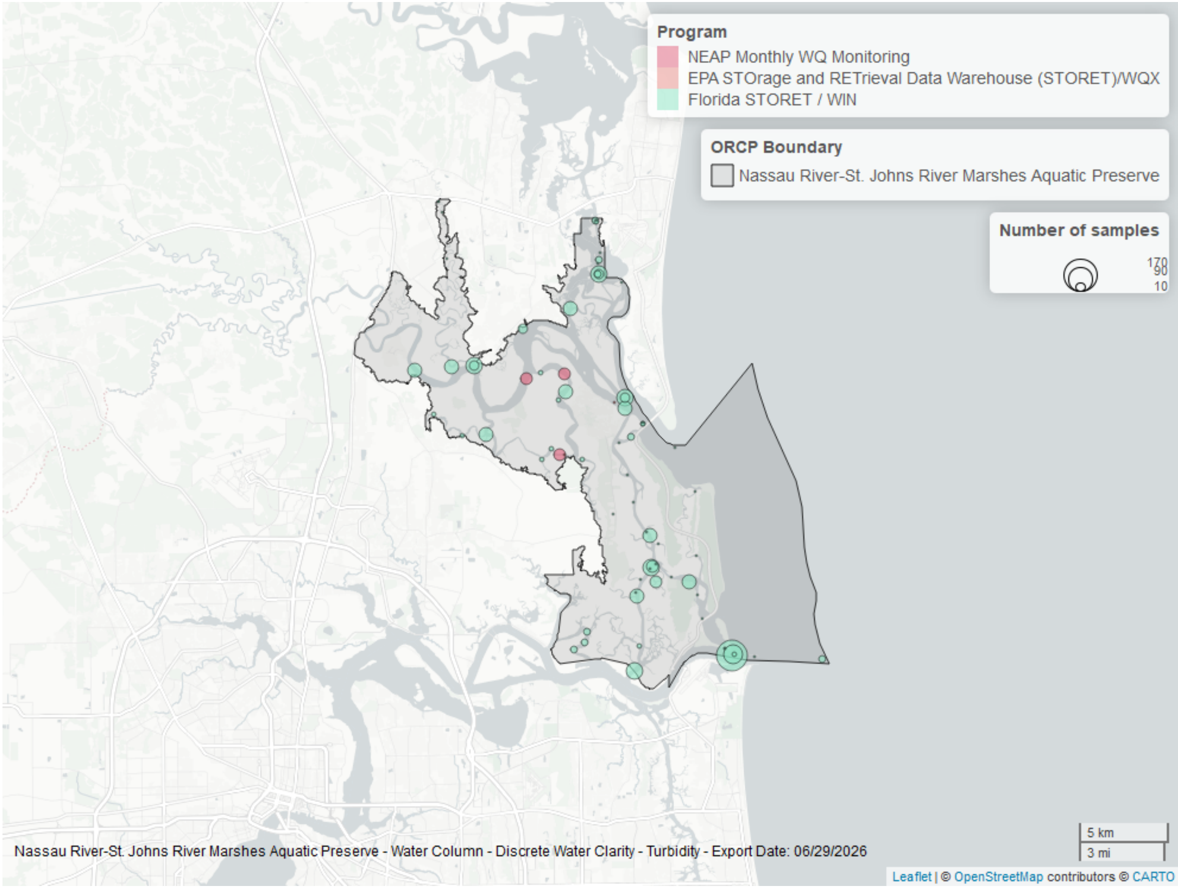

Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

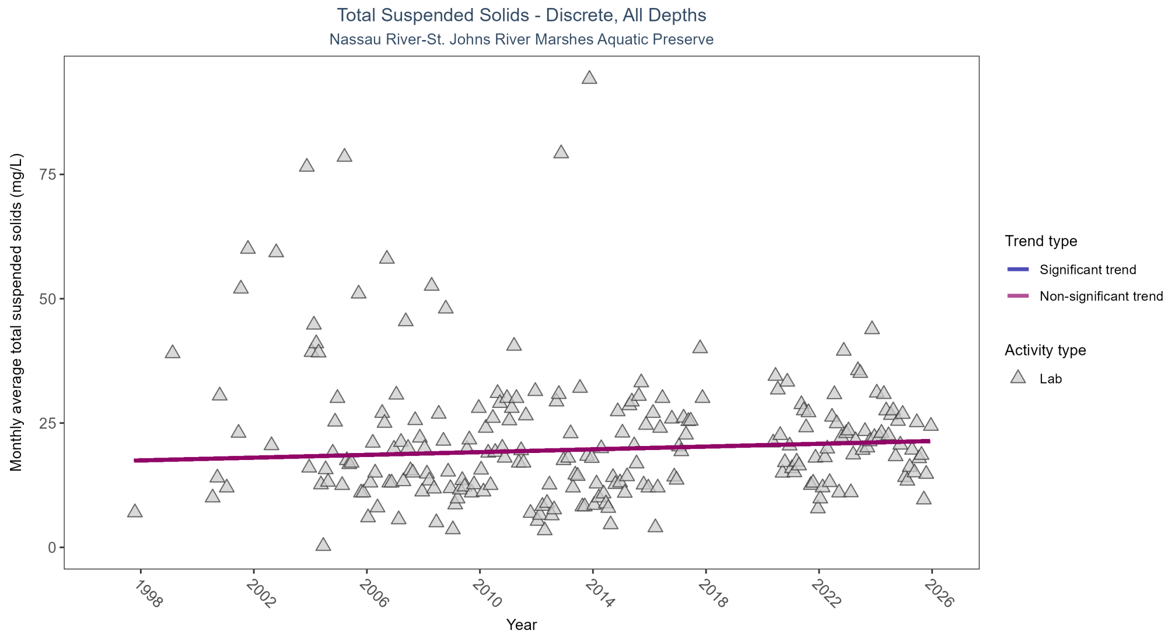

Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 15: Total suspended solids showed no detectable trend between 1997 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	965	26	1997 - 2025	18	0.0678	17.34193	0.1392	0.1502

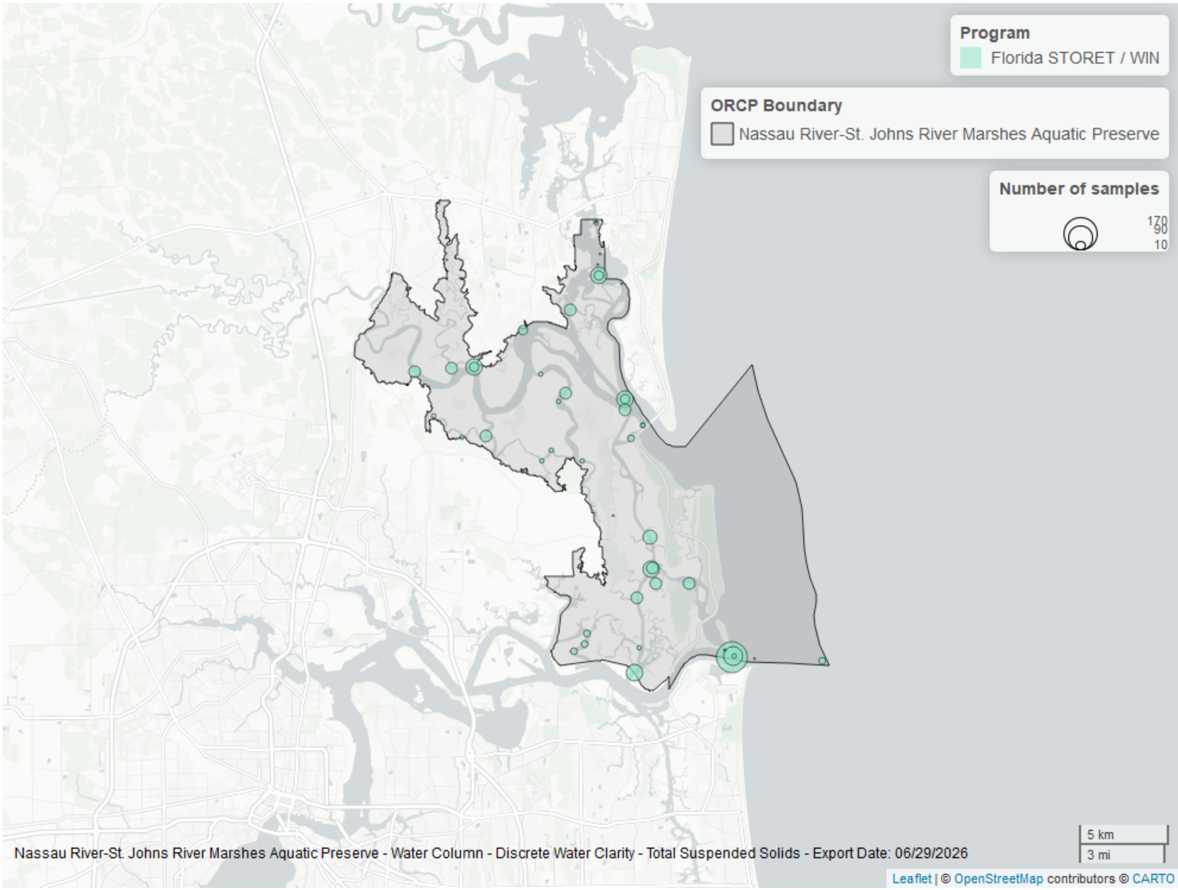

Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

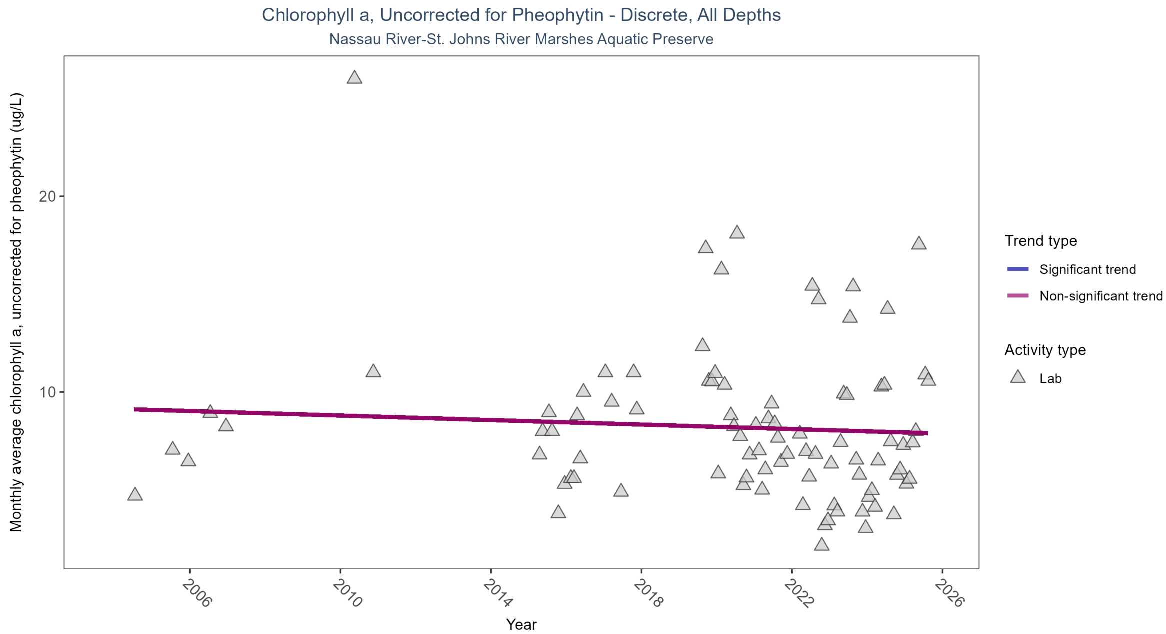

Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 16: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2004 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	405	14	2004 - 2025	6.8	-0.10118	9.14587	-0.05758	0.513

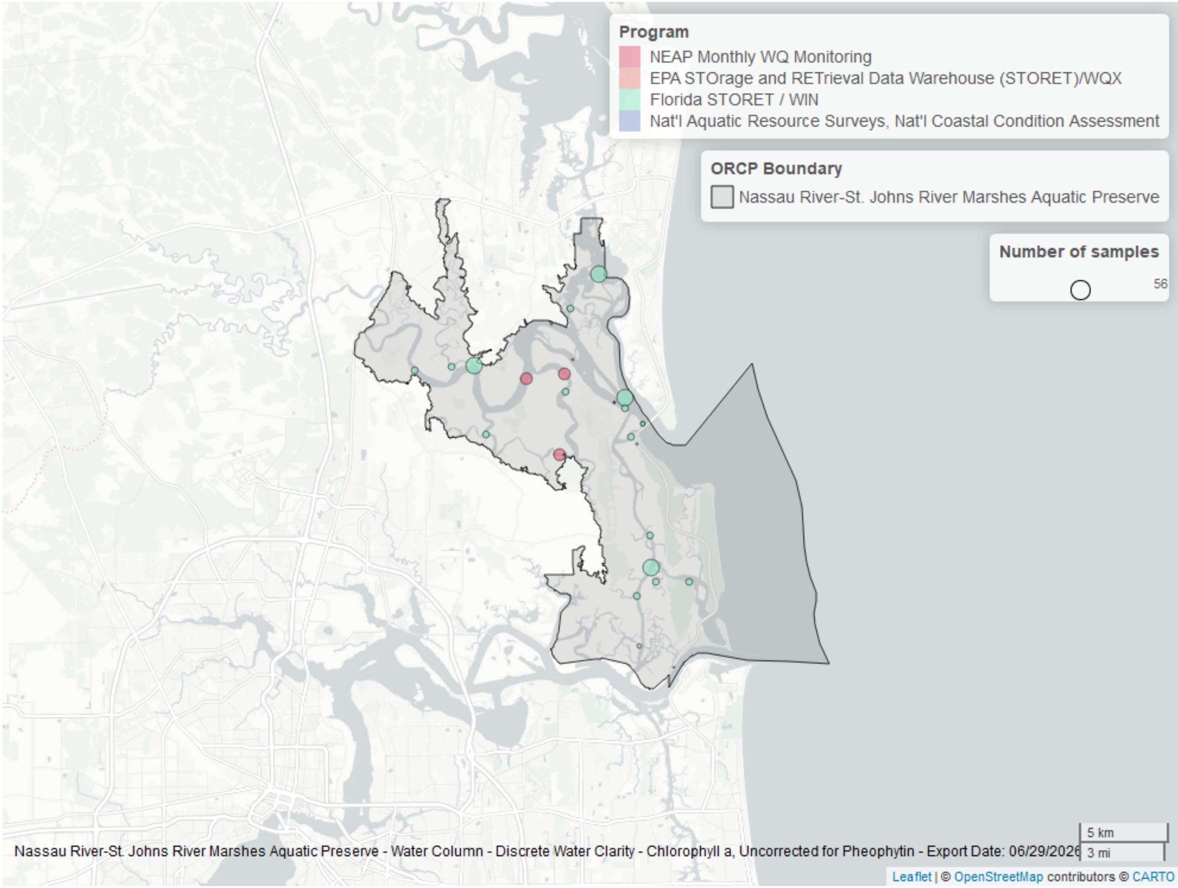

Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

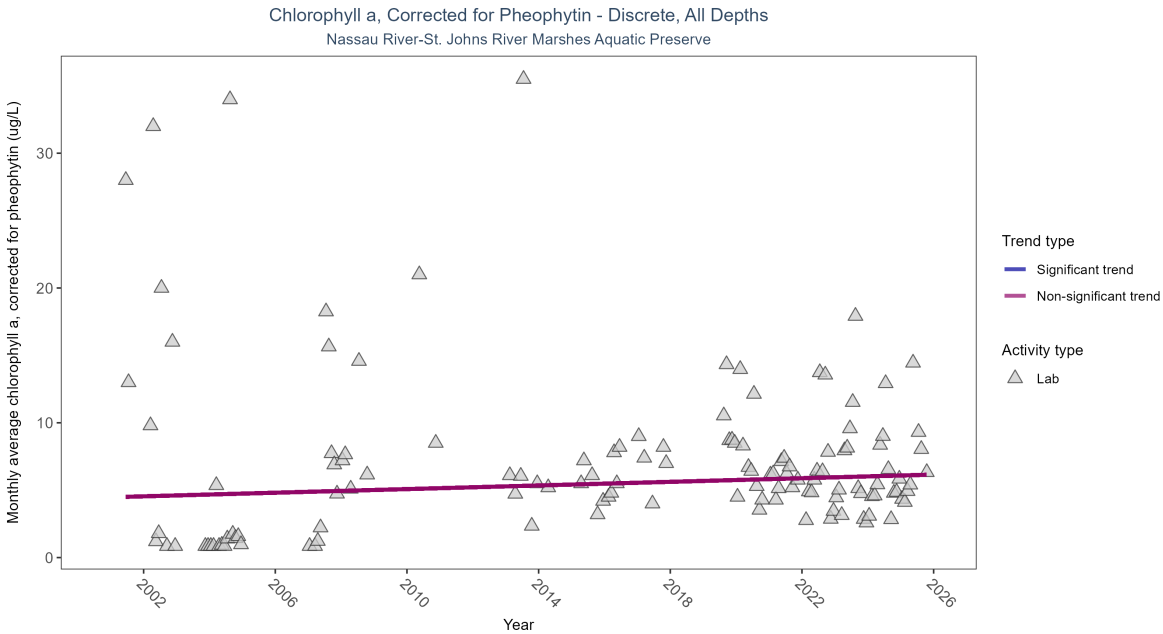

Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2001 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	756	19	2001 - 2025	5.7	0.10262	4.47525	0.06721	0.1212

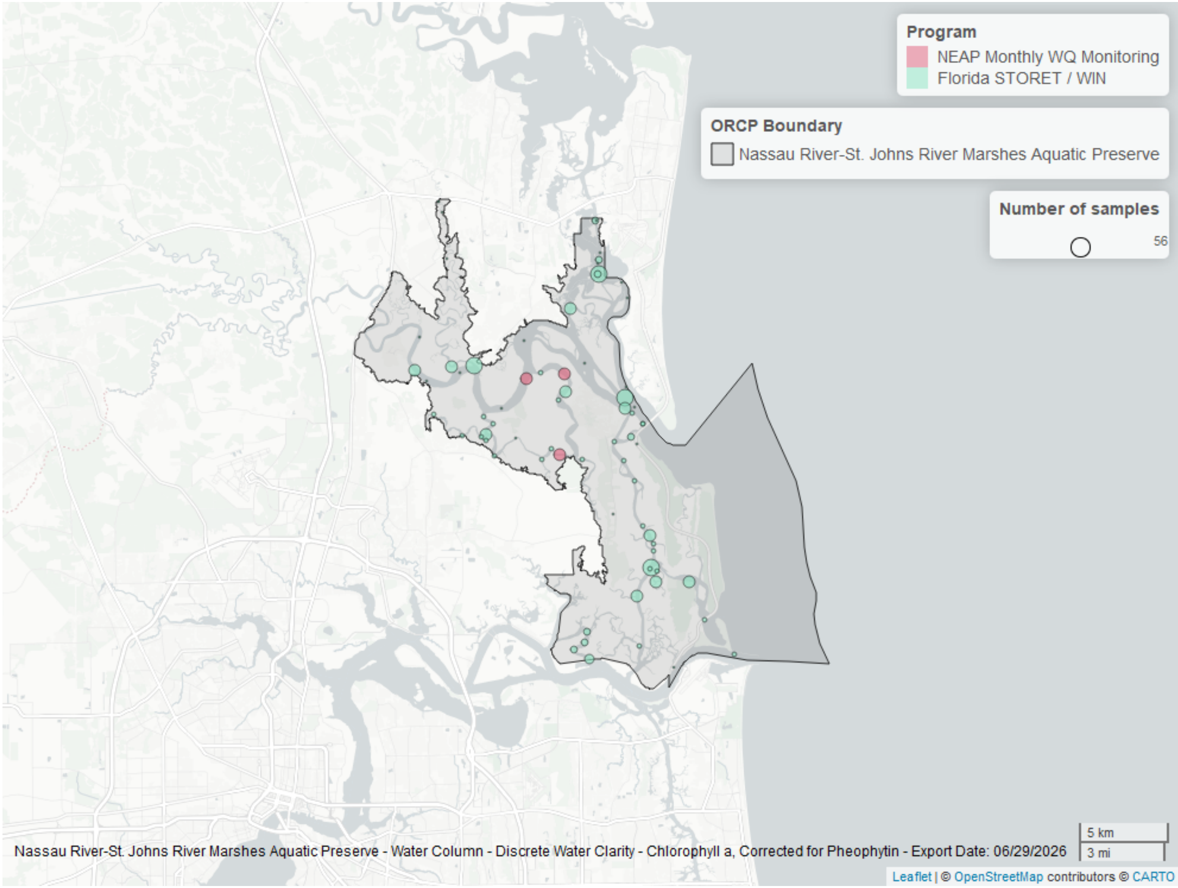

Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

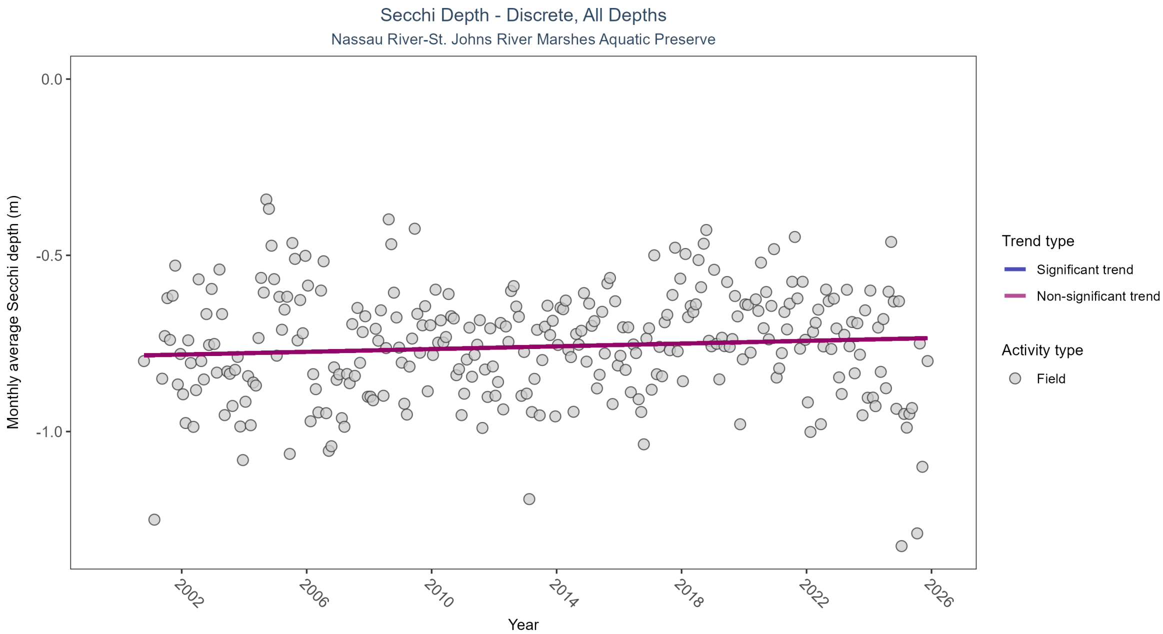

Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 18: Secchi depth showed no detectable trend between 2000 and 2025.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	20451	26	2000 - 2025	-0.7	0.06628	-0.78549	0.00195	0.1168

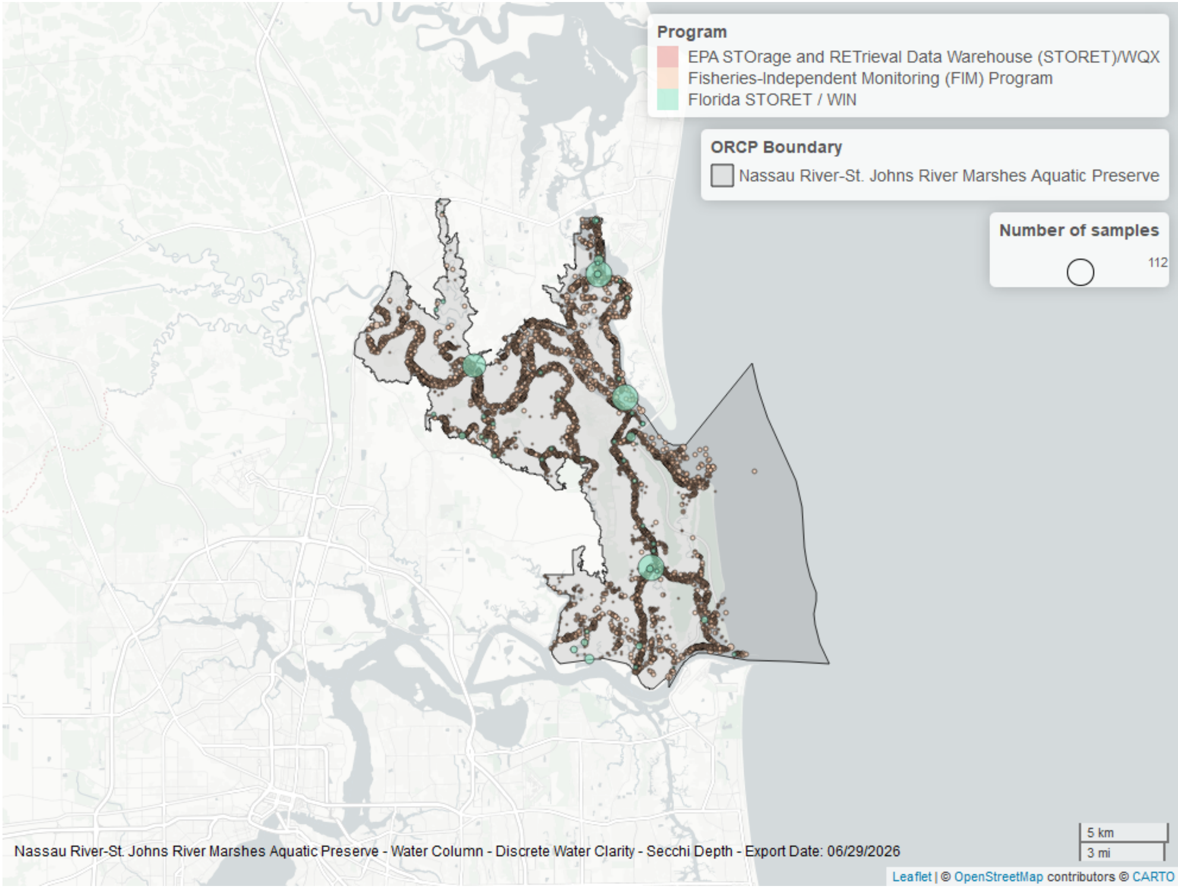

Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

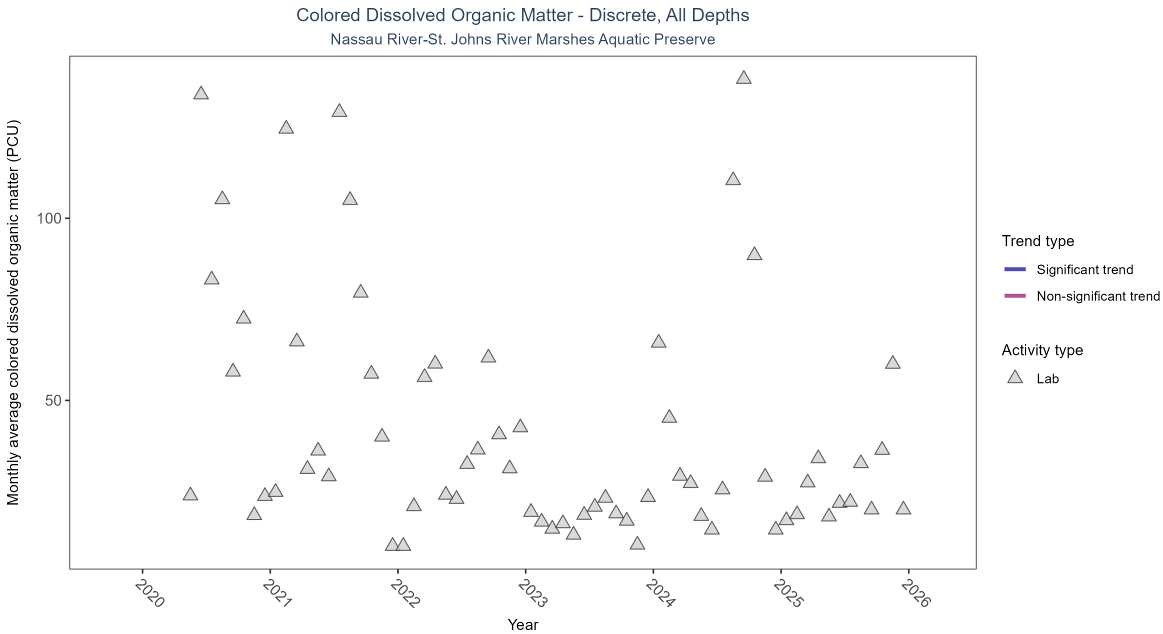

Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 19: There was insufficient data to fit a model for colored dissolved organic matter.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	588	6	2020 - 2025	24.53637	-	-	-	-

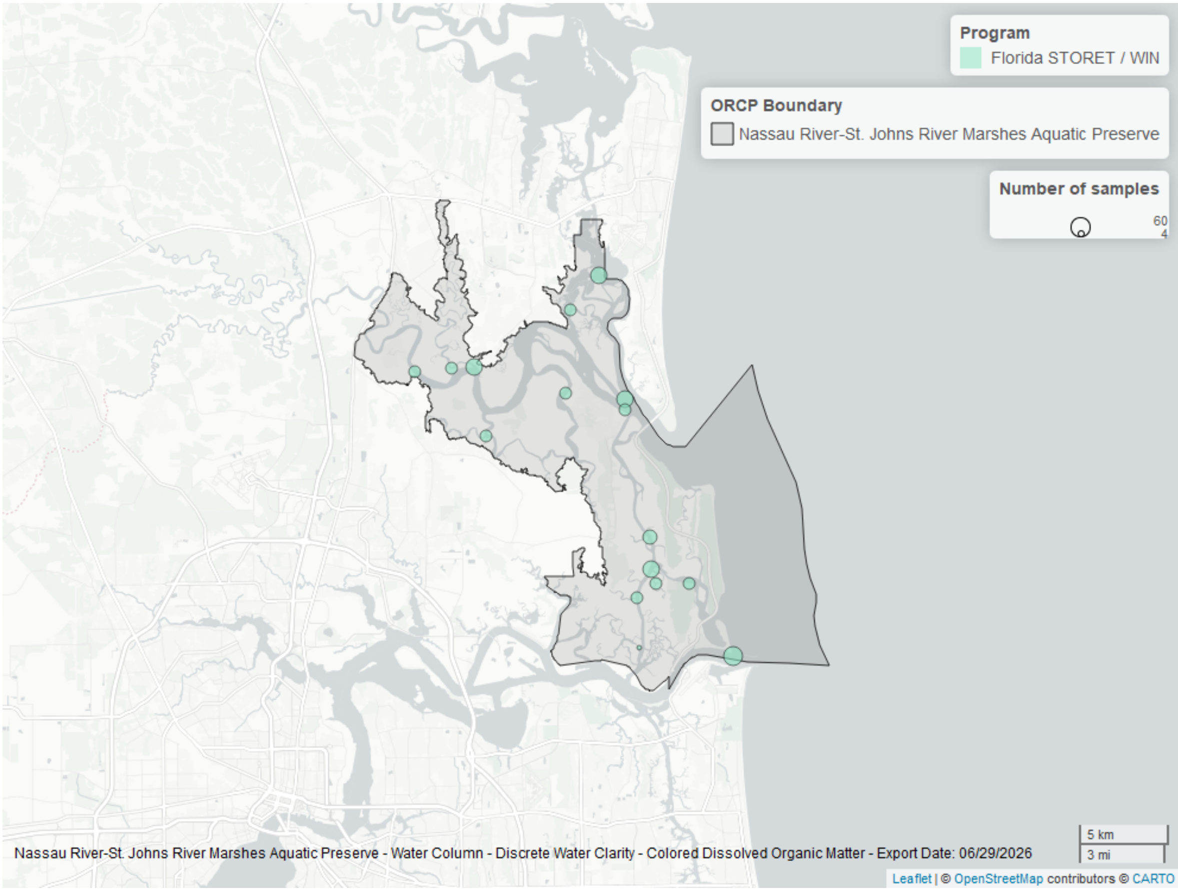

Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.